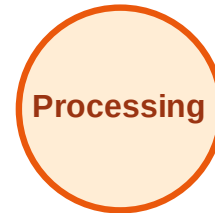


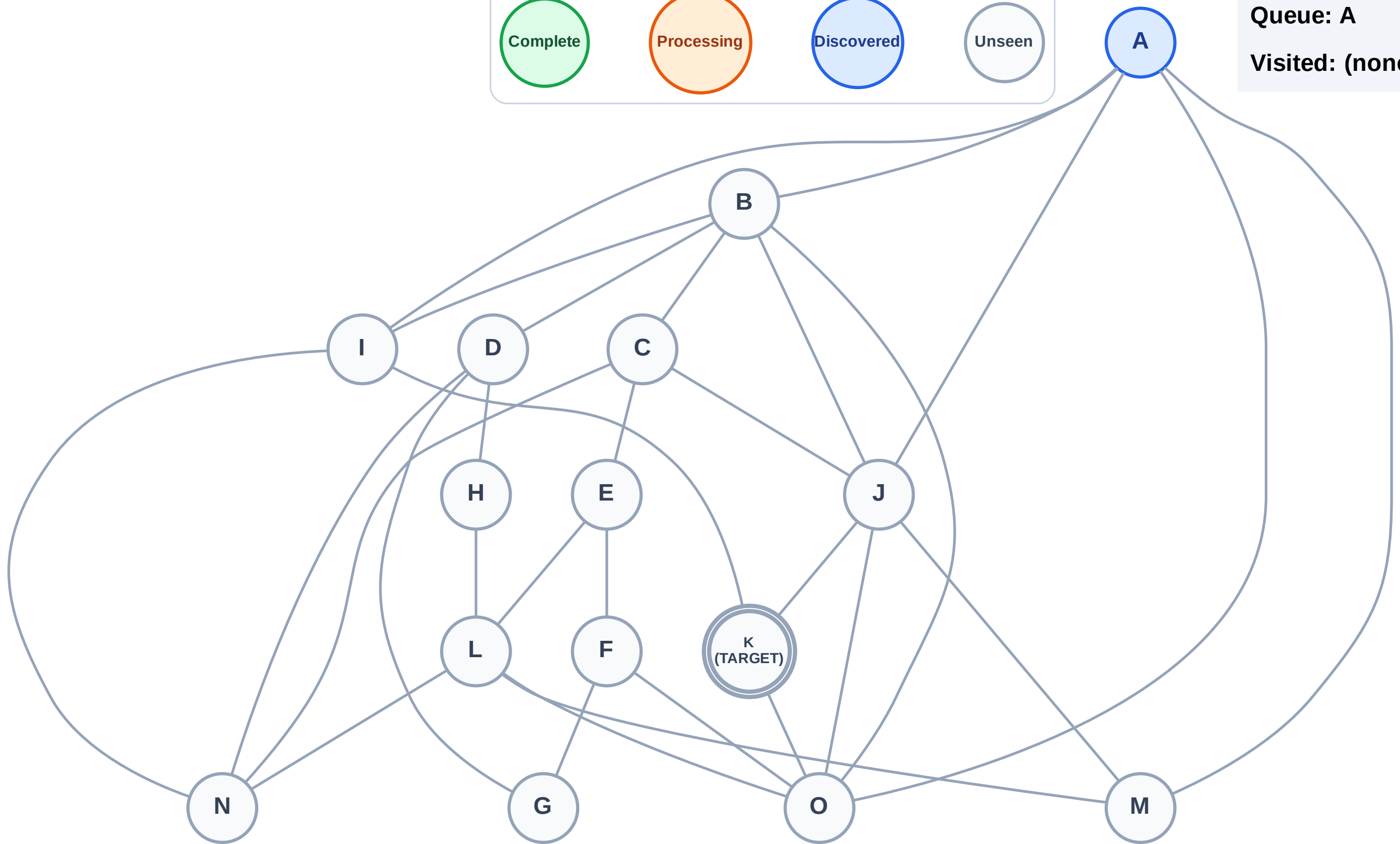
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



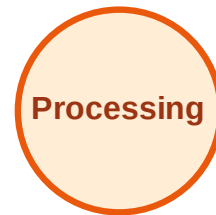
Queue: A
Visited: (none)



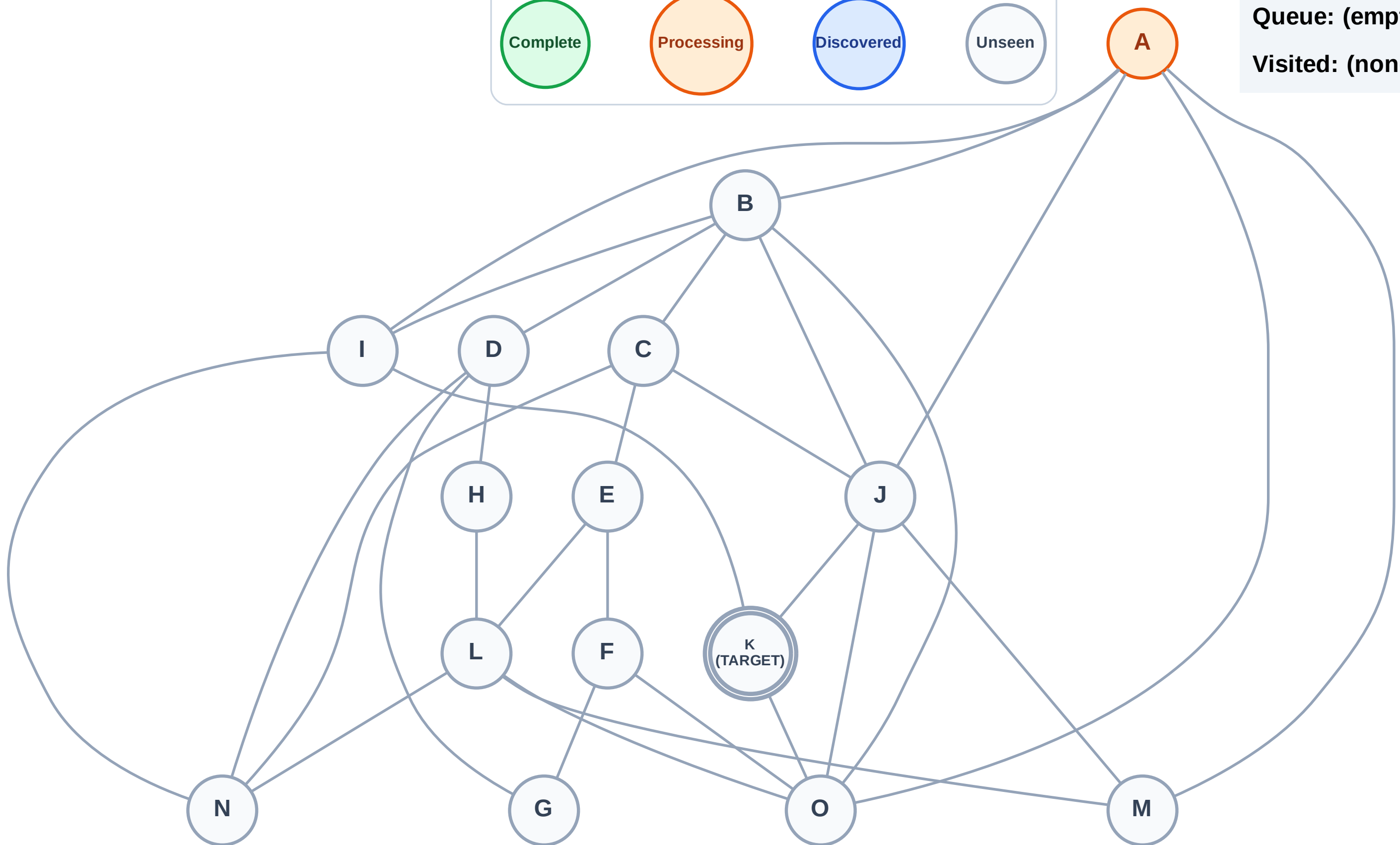
BFS Traversal

Step 2: Process node A

Legend



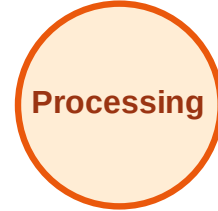
Queue: (empty)
Visited: (none)



BFS Traversal

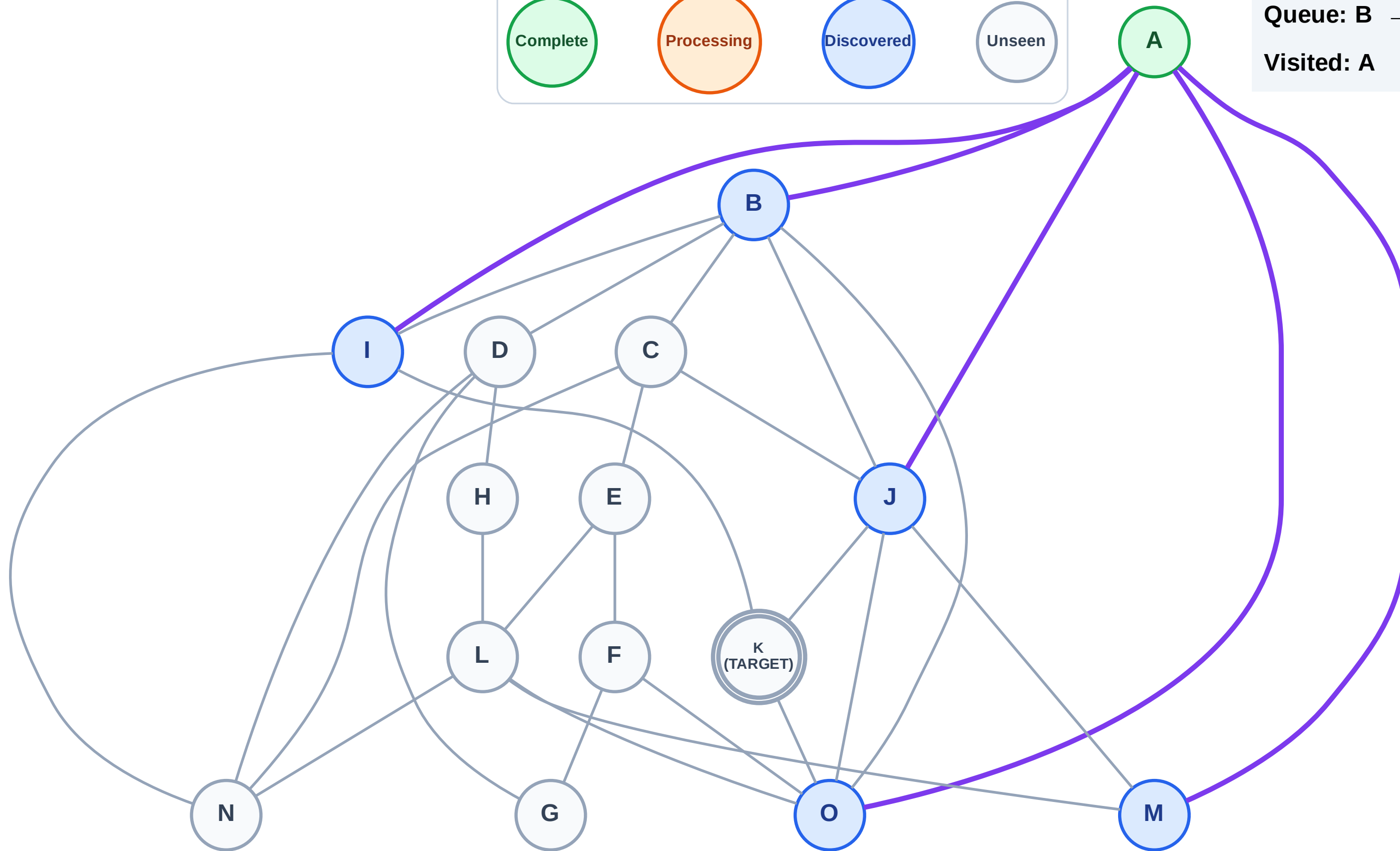
Step 3: Discover B, I, J, M, O from A

Legend



Queue: B → I → J → M → O

Visited: A



BFS Traversal

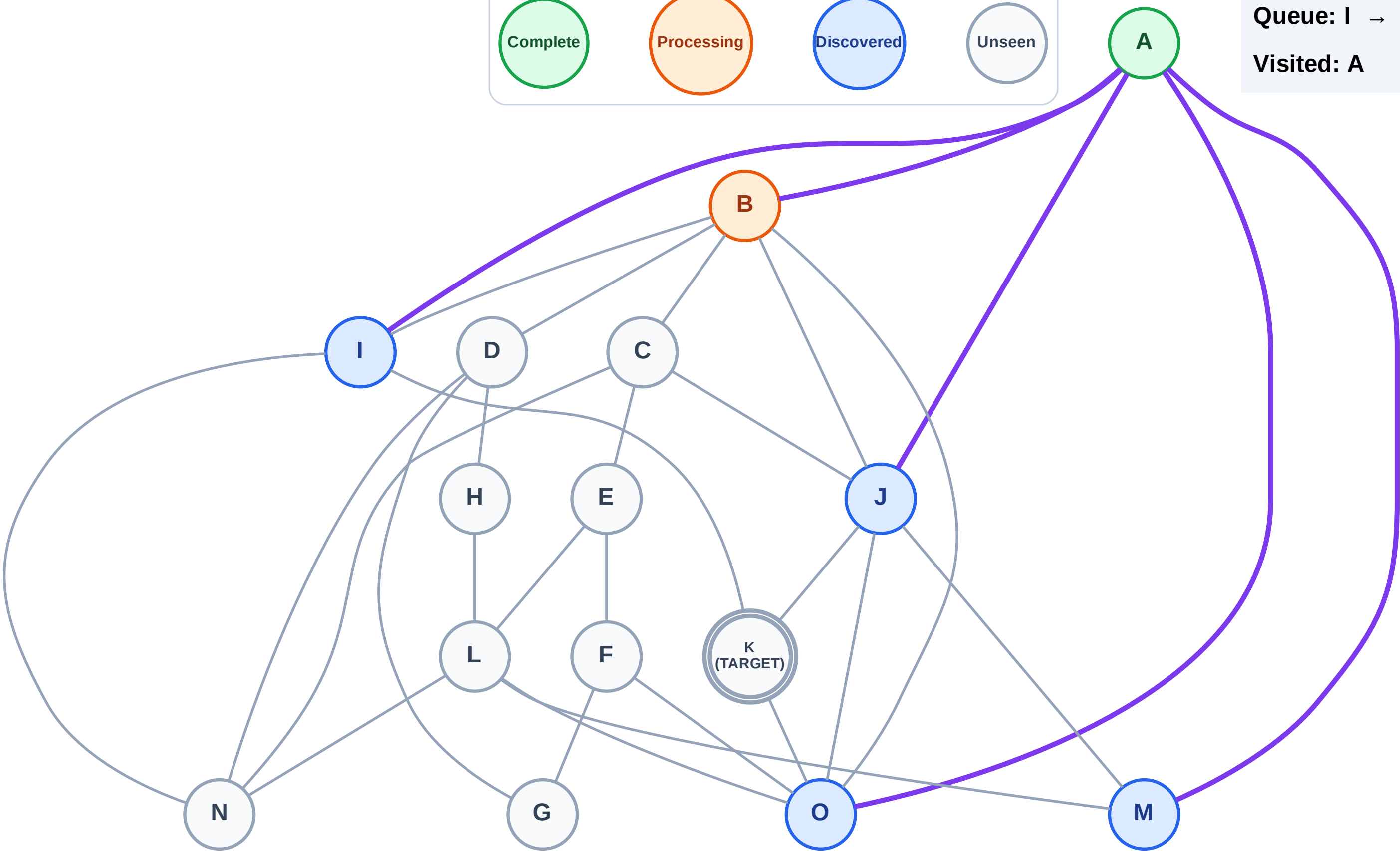
Step 4: Process node B

Legend



Queue: I → J → M → O

Visited: A



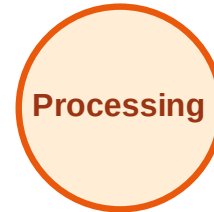
BFS Traversal

Step 5: Discover C, D from B

Legend



Complete



Processing



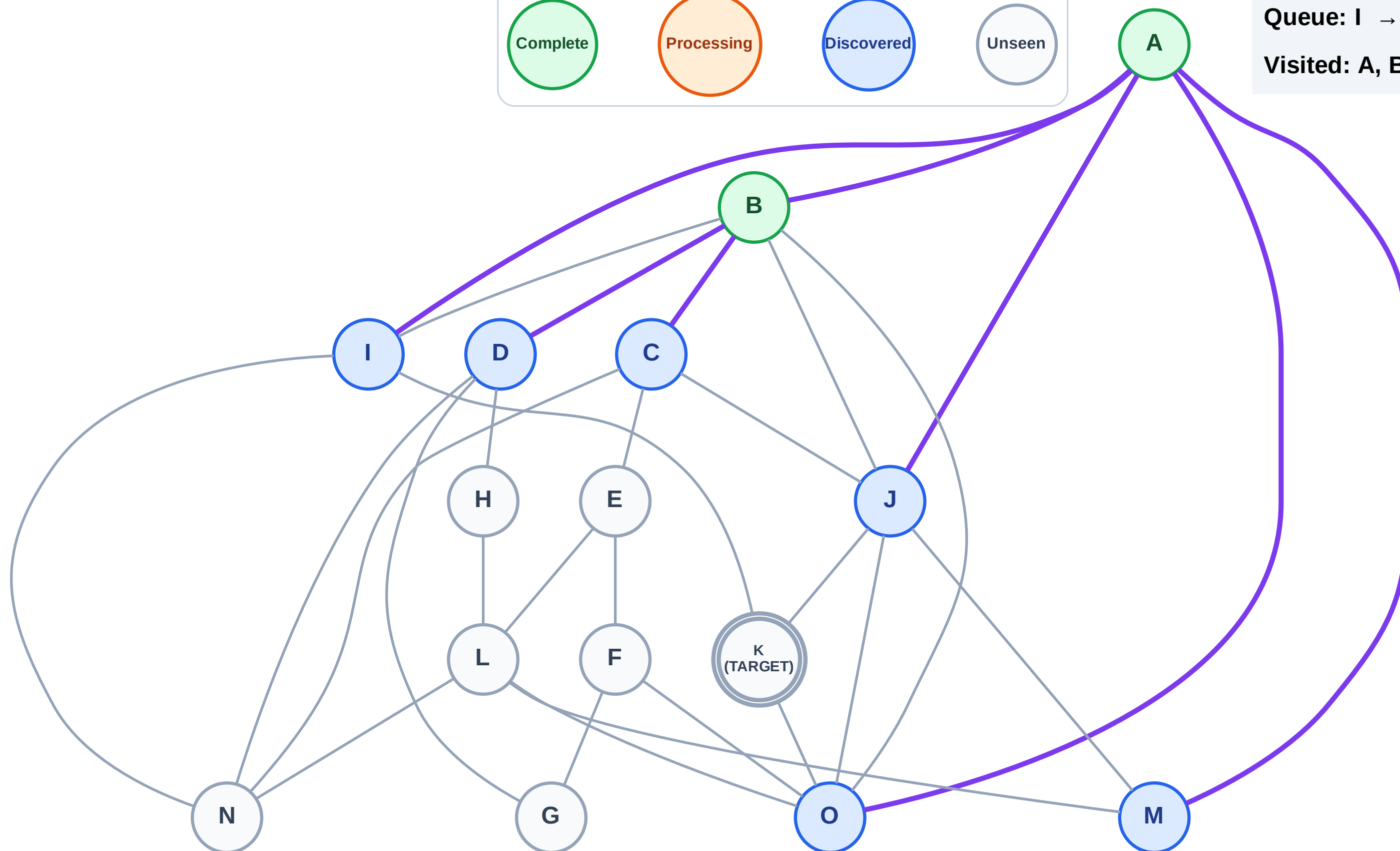
Discovered



Unseen

Queue: I → J → M → O → C → D

Visited: A, B



BFS Traversal

Step 6: Process node I

Legend

Complete

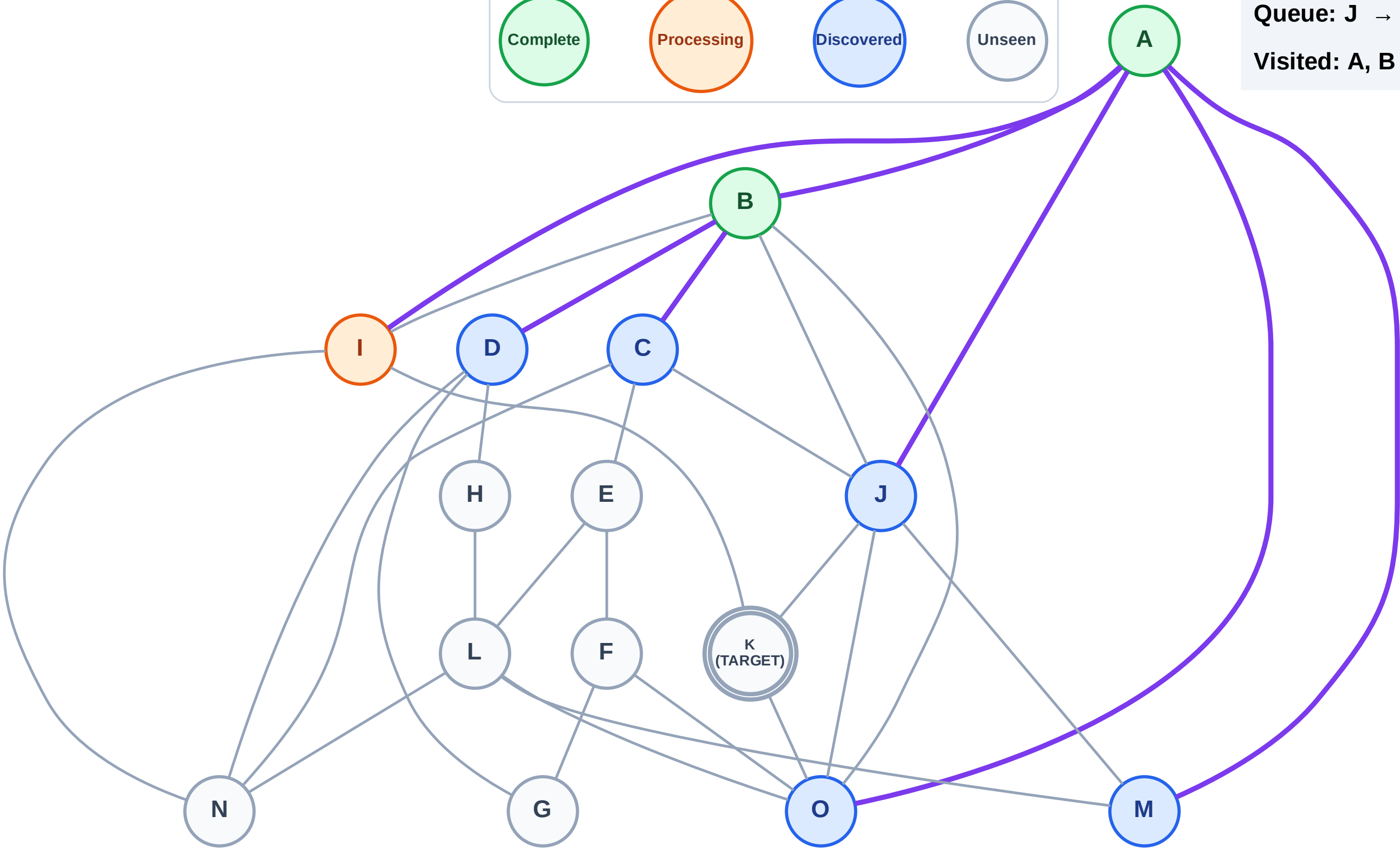
Processing

Discovered

Unseen

Queue: J → M → O → C → D

Visited: A, B

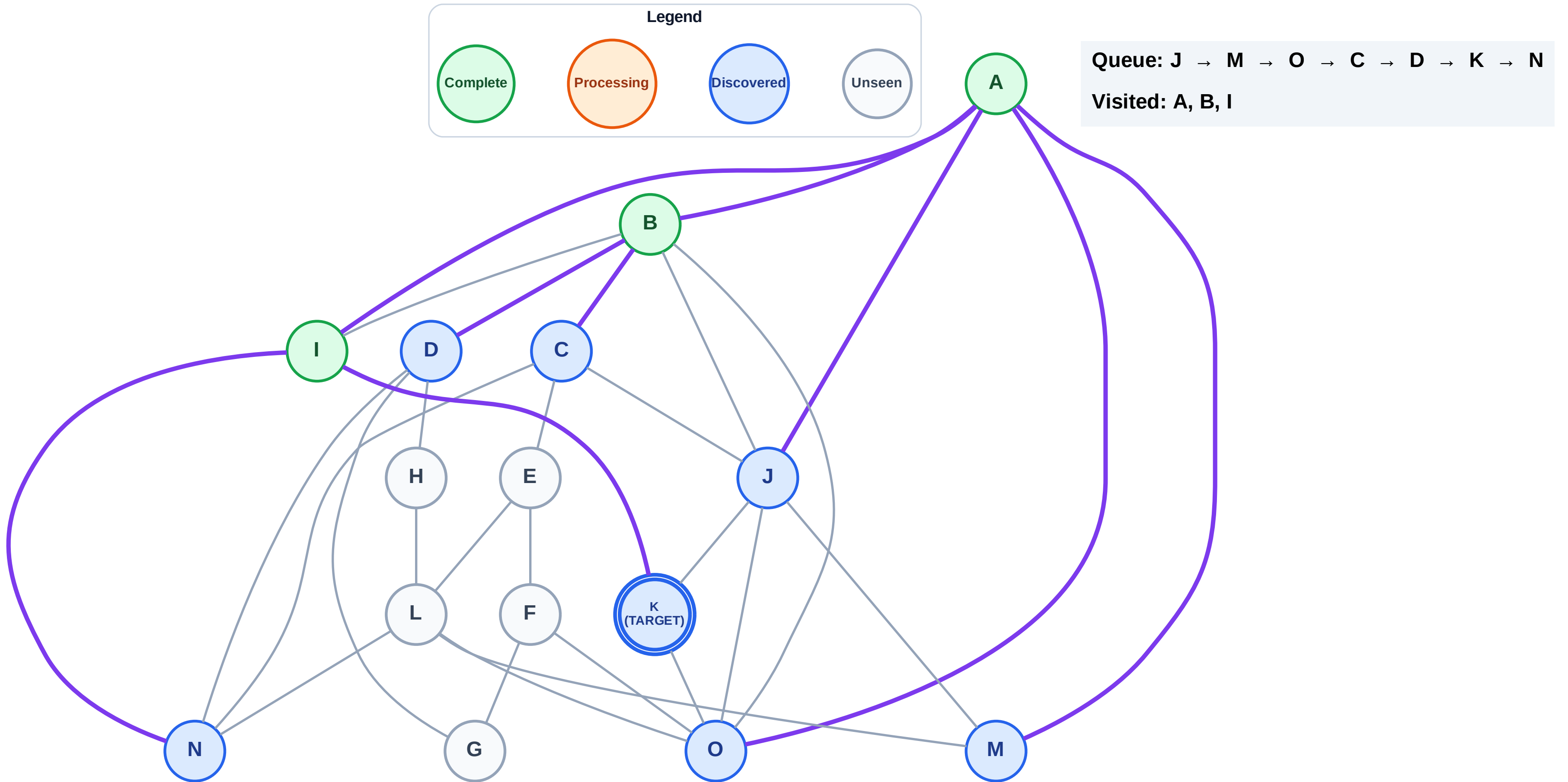


Step 7: Discover K, N from I

Legend

Unseen

Visited: A, B, I



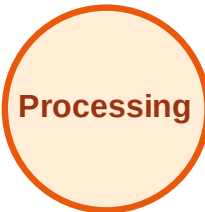
BFS Traversal

Step 8: Process node J

Legend



Complete



Processing



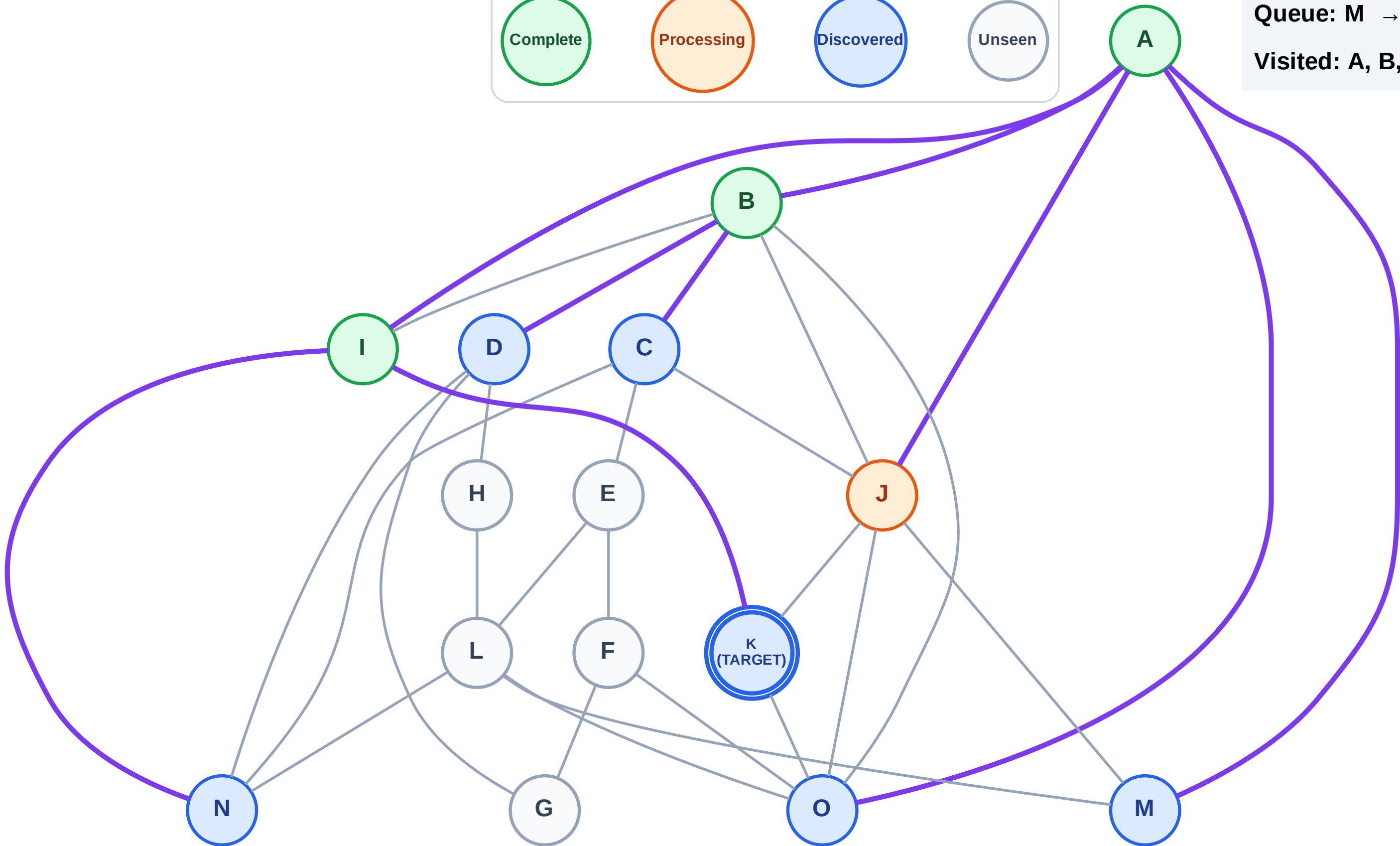
Discovered



Unseen

Queue: M → O → C → D → K → N

Visited: A, B, I



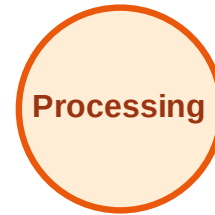
BFS Traversal

Step 9: No new neighbors from J

Legend



Complete



Processing



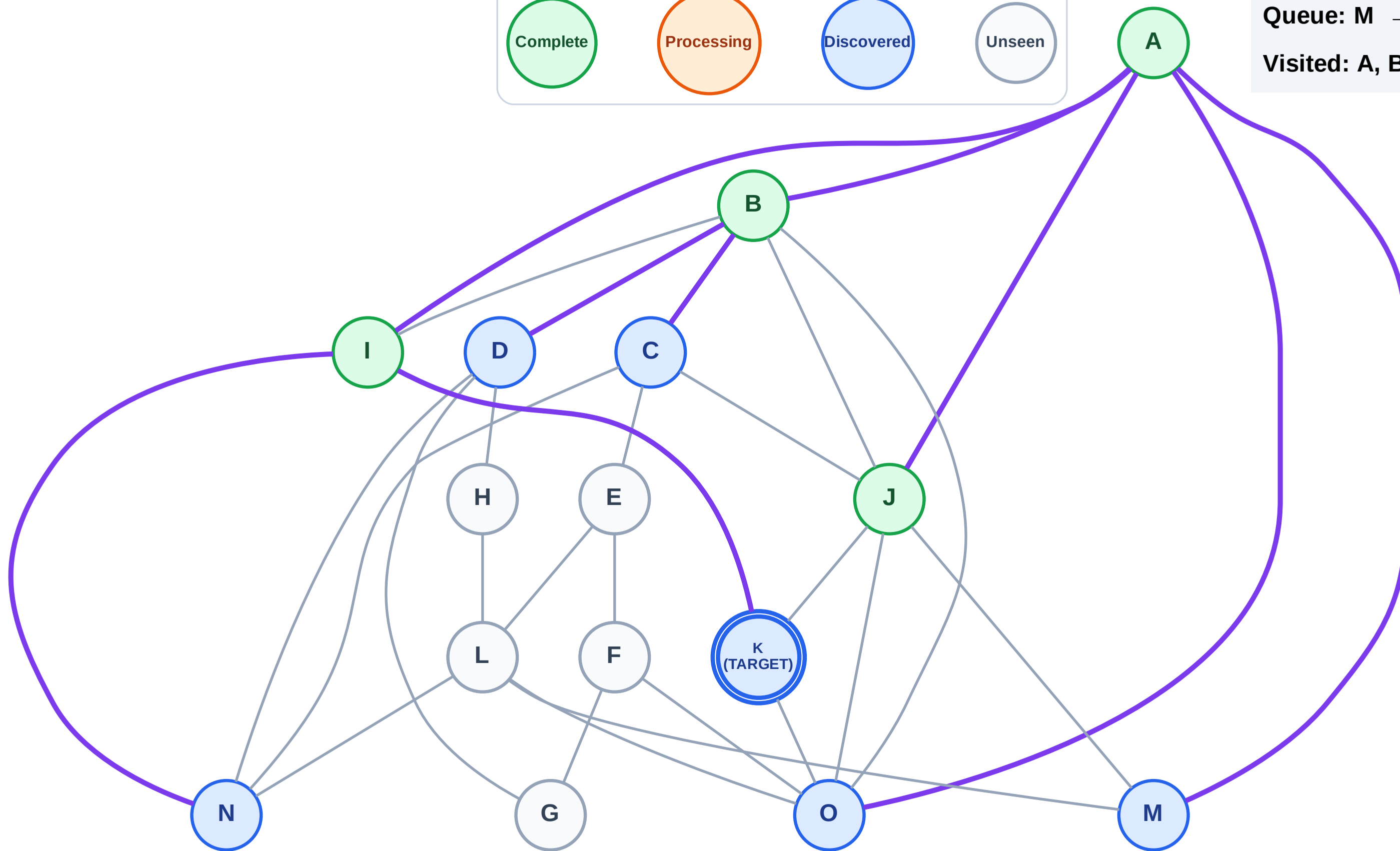
Discovered



Unseen

Queue: M → O → C → D → K → N

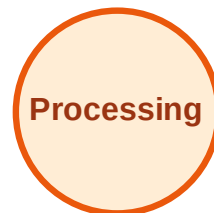
Visited: A, B, I, J



BFS Traversal

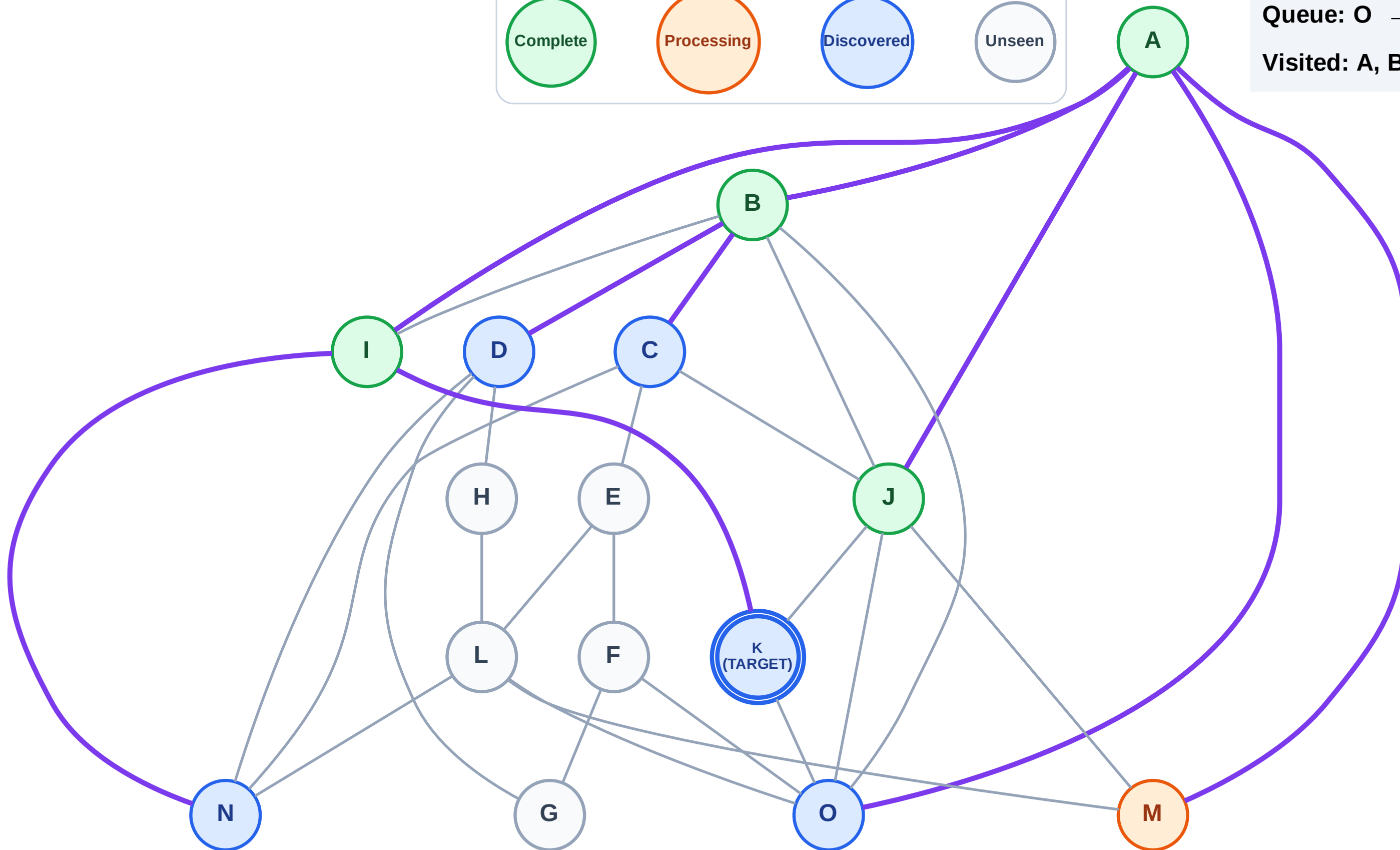
Step 10: Process node M

Legend



Queue: O → C → D → K → N

Visited: A, B, I, J



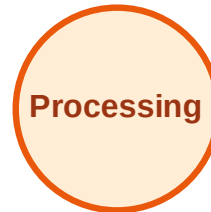
BFS Traversal

Step 11: Discover L from M

Legend



Complete



Processing



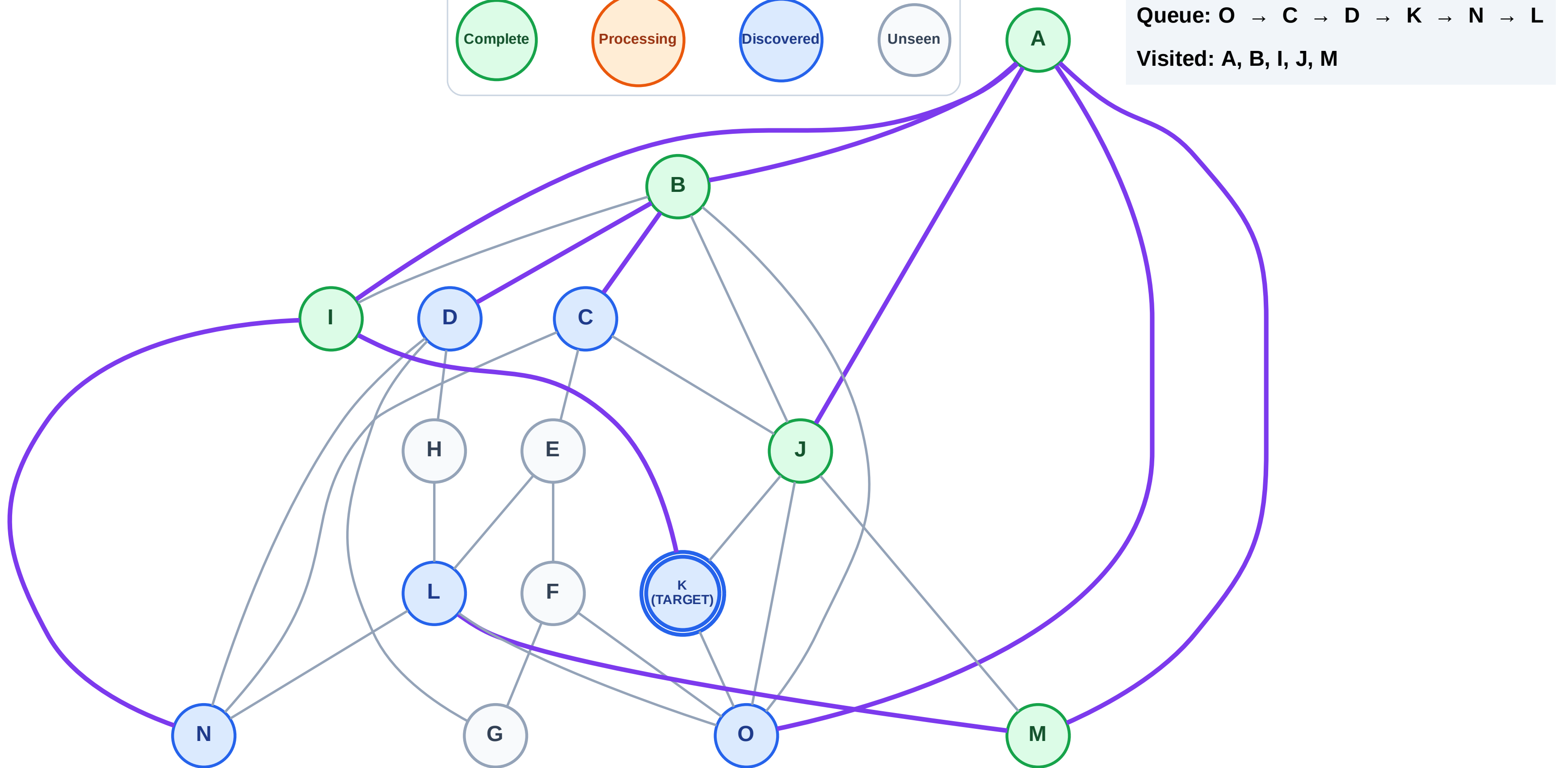
Discovered



Unseen

Queue: O → C → D → K → N → L

Visited: A, B, I, J, M



BFS Traversal

Step 12: Process node O

Legend

Complete

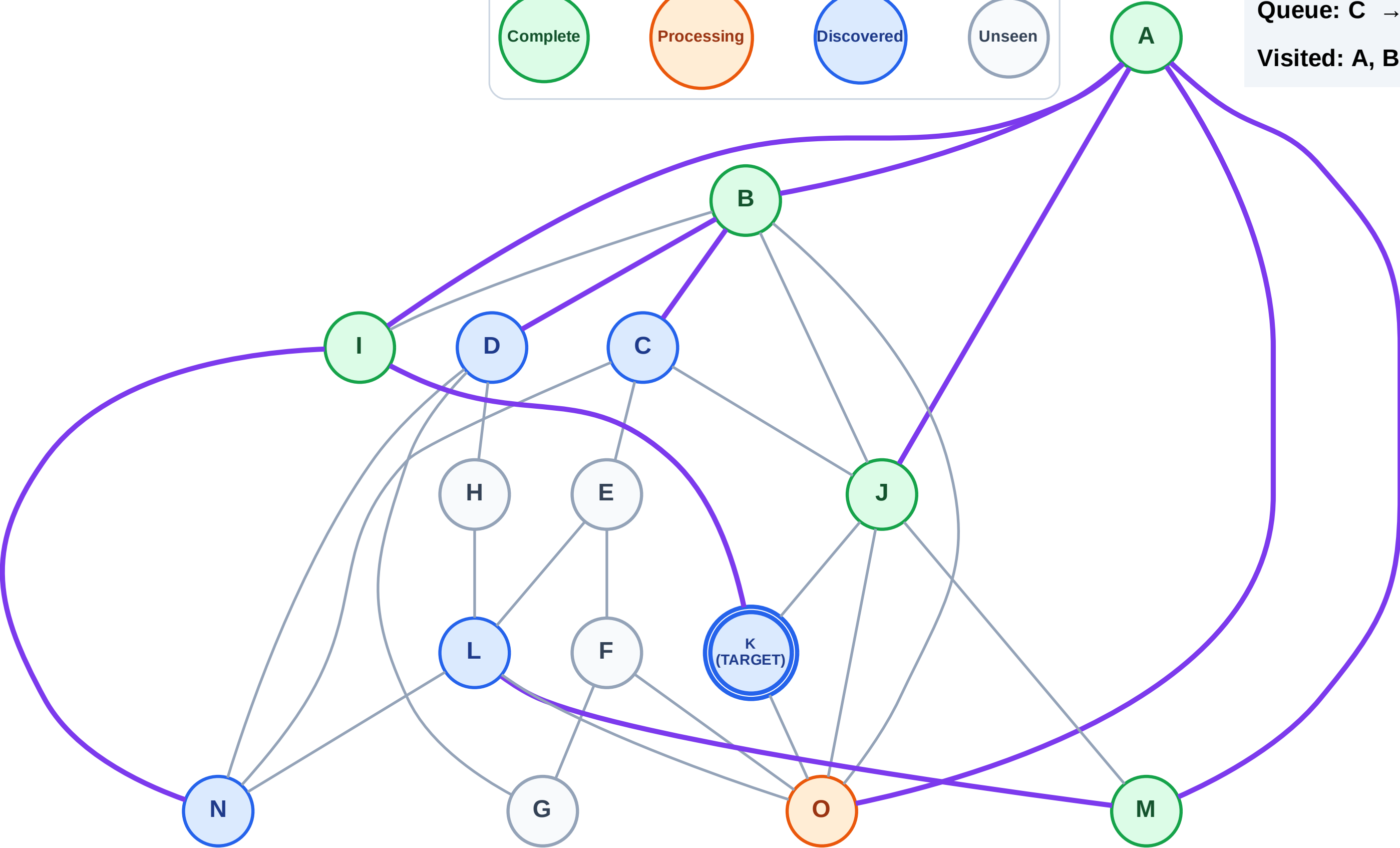
Processing

Discovered

Unseen

Queue: C → D → K → N → L

Visited: A, B, I, J, M



BFS Traversal

Step 13: Discover F from O

Legend



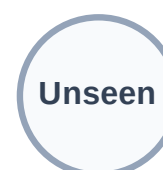
Complete



Processing



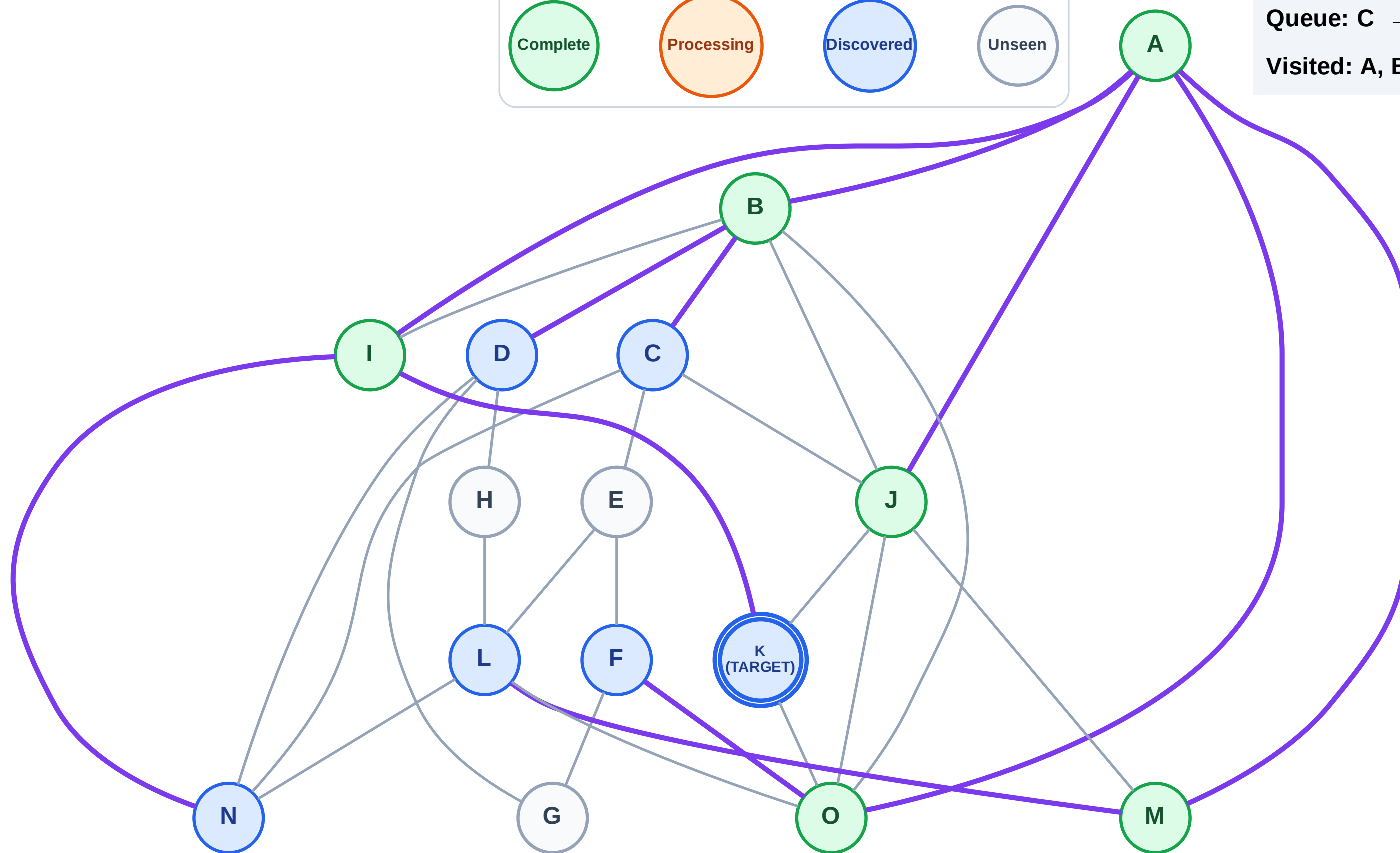
Discovered



Unseen

Queue: C → D → K → N → L → F

Visited: A, B, I, J, M, O



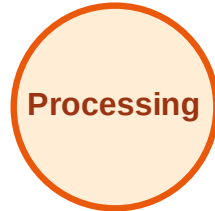
BFS Traversal

Step 14: Process node C

Legend



Complete



Processing



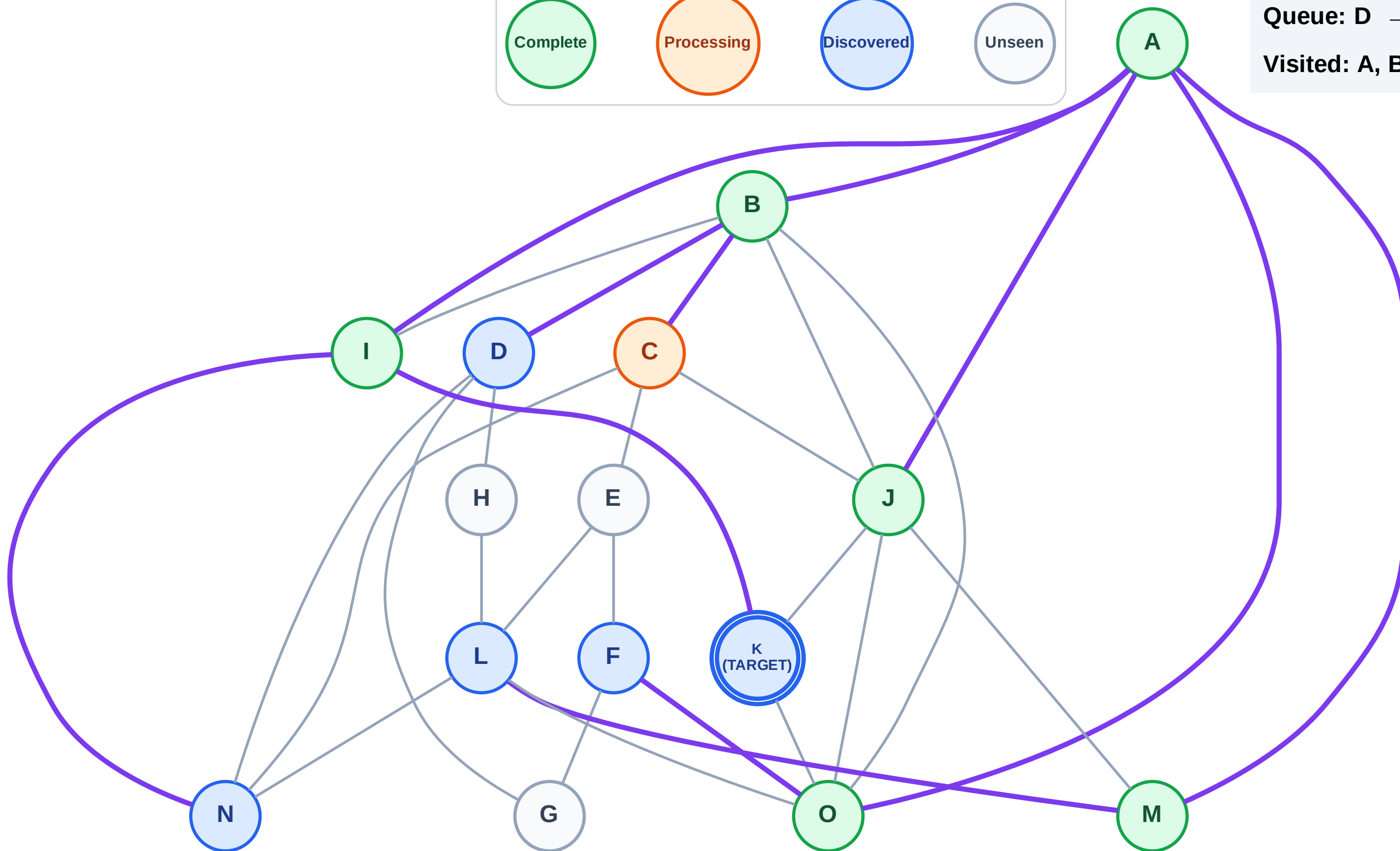
Discovered



Unseen

Queue: D → K → N → L → F

Visited: A, B, I, J, M, O



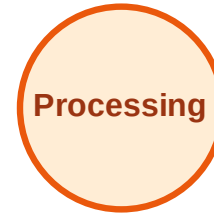
BFS Traversal

Step 15: Discover E from C

Legend



Complete



Processing



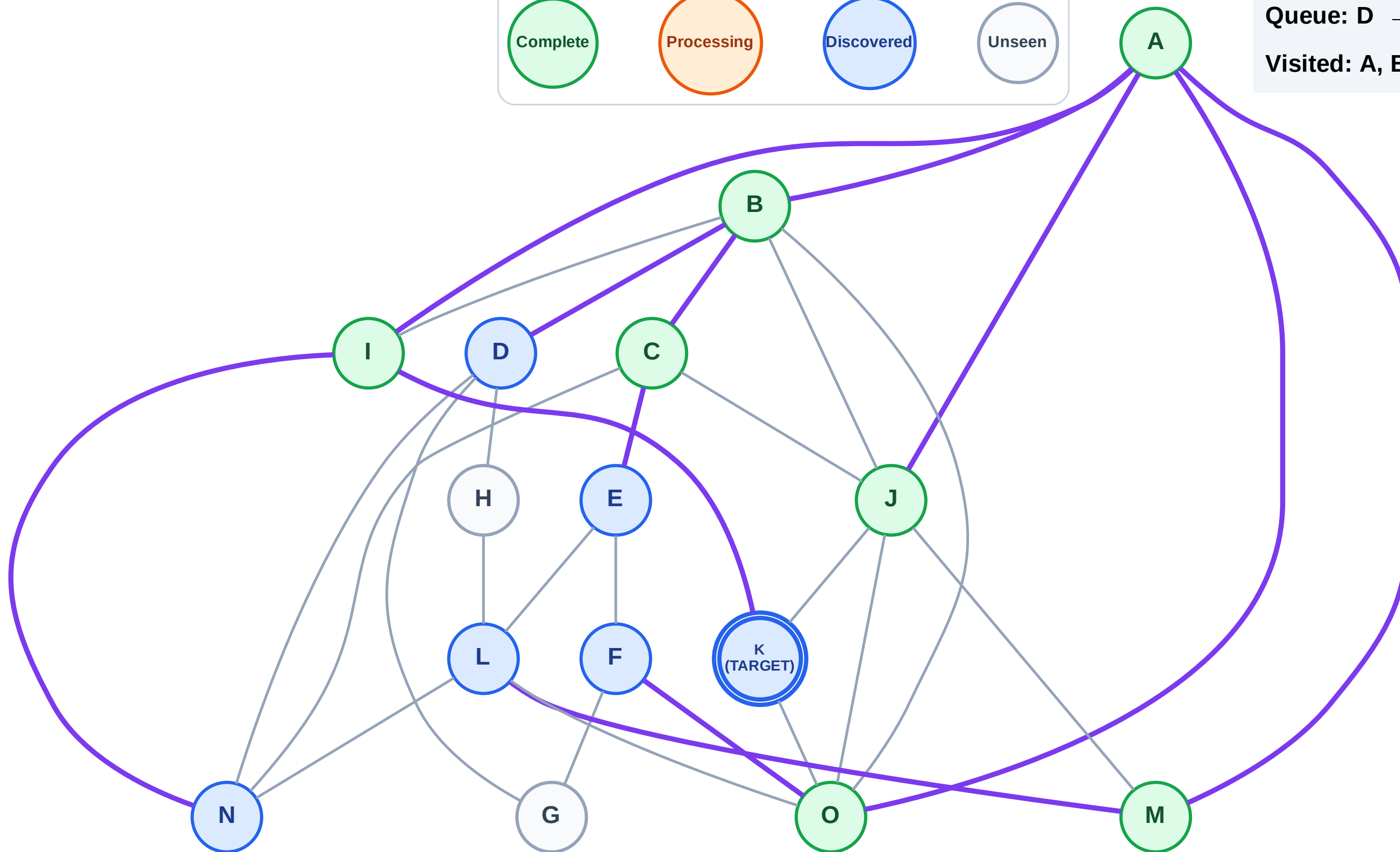
Discovered



Unseen

Queue: D → K → N → L → F → E

Visited: A, B, C, I, J, M, O



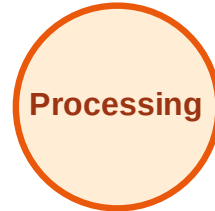
BFS Traversal

Step 16: Process node D

Legend



Complete



Processing



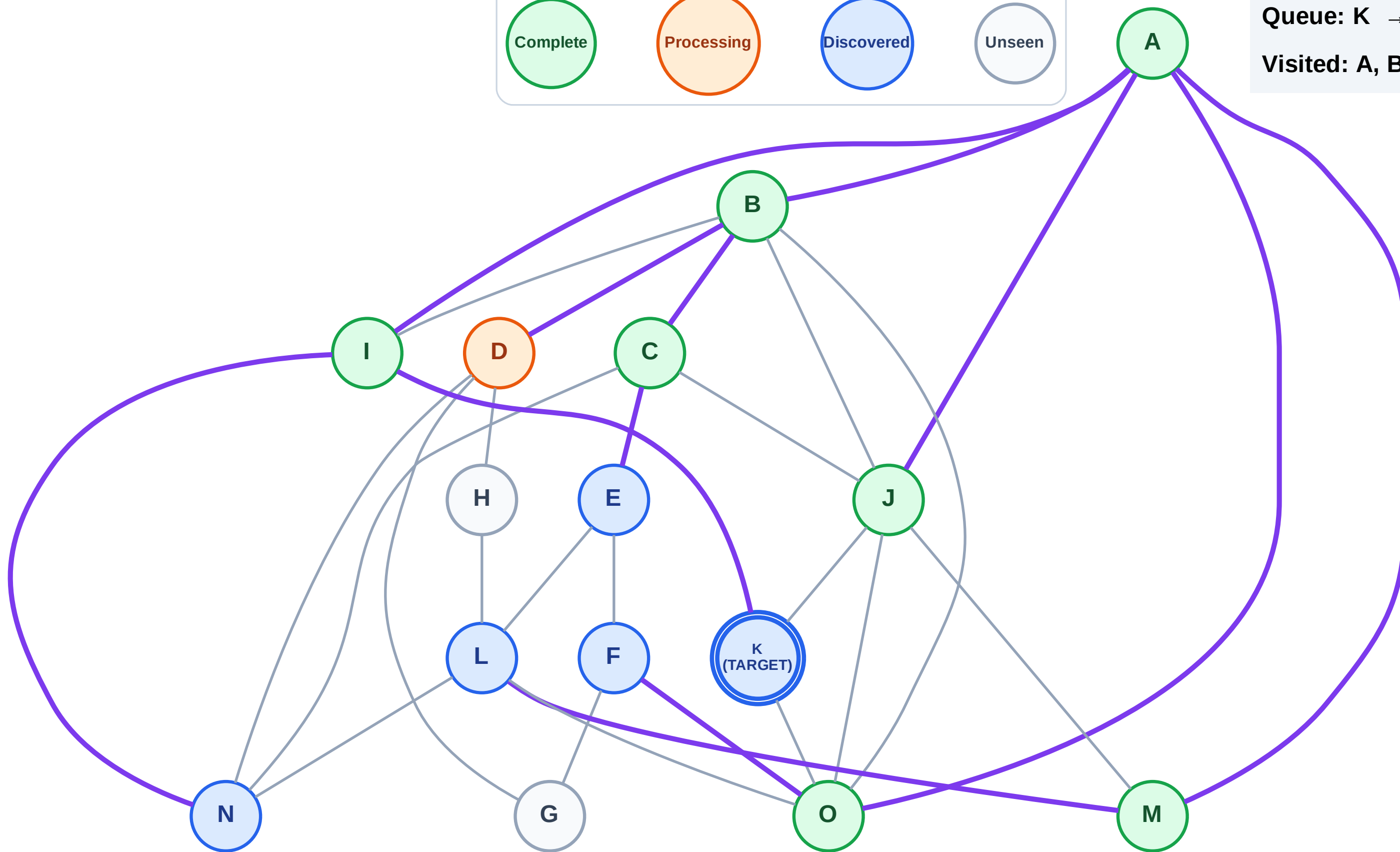
Discovered



Unseen

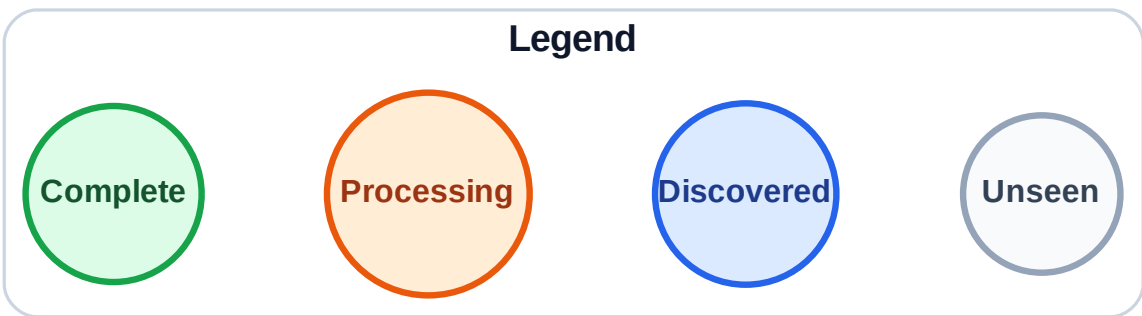
Queue: K → N → L → F → E

Visited: A, B, C, I, J, M, O



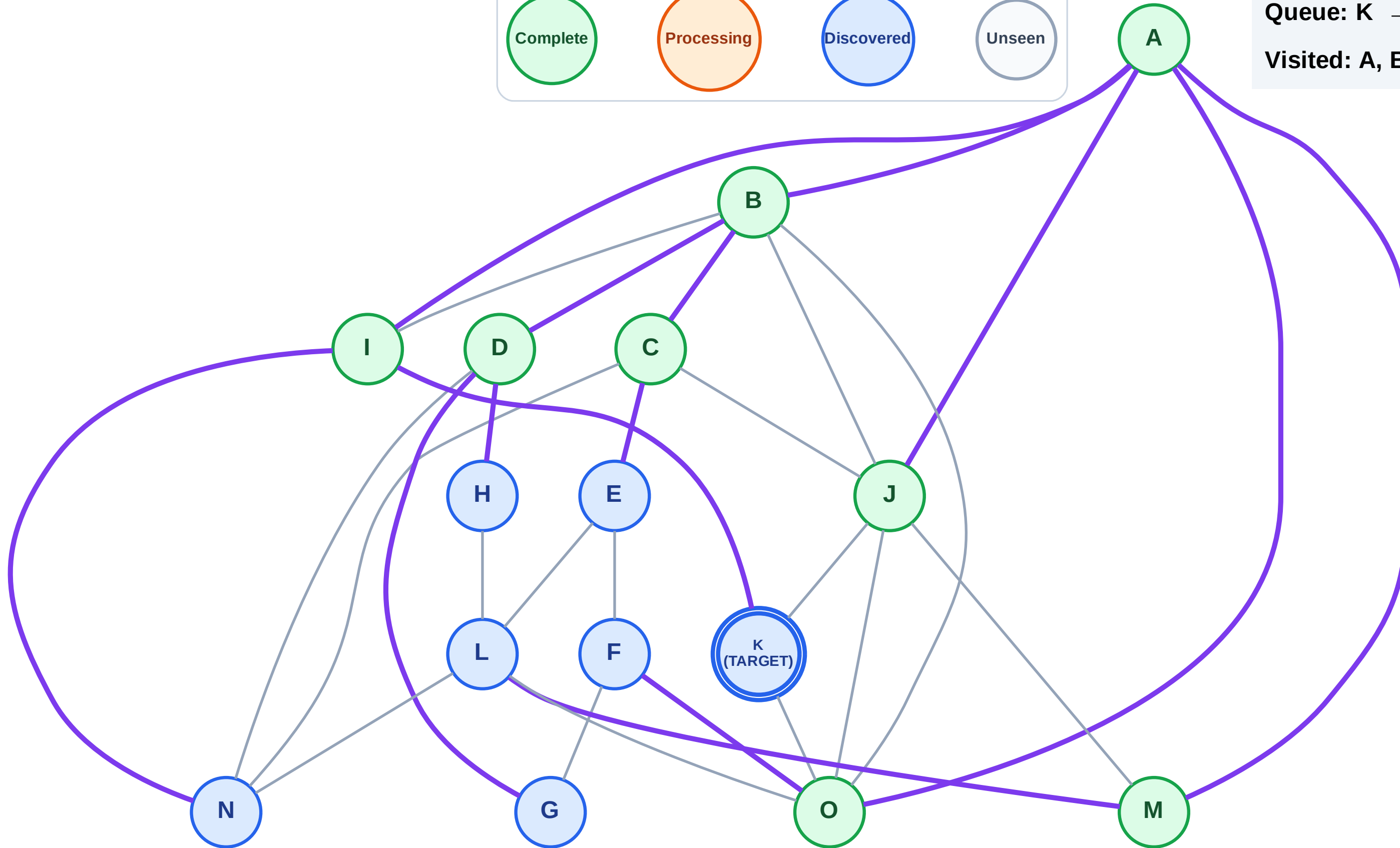
BFS Traversal

Step 17: Discover G, H from D



Queue: K → N → L → F → E → G → H

Visited: A, B, C, D, I, J, M, O



BFS Traversal

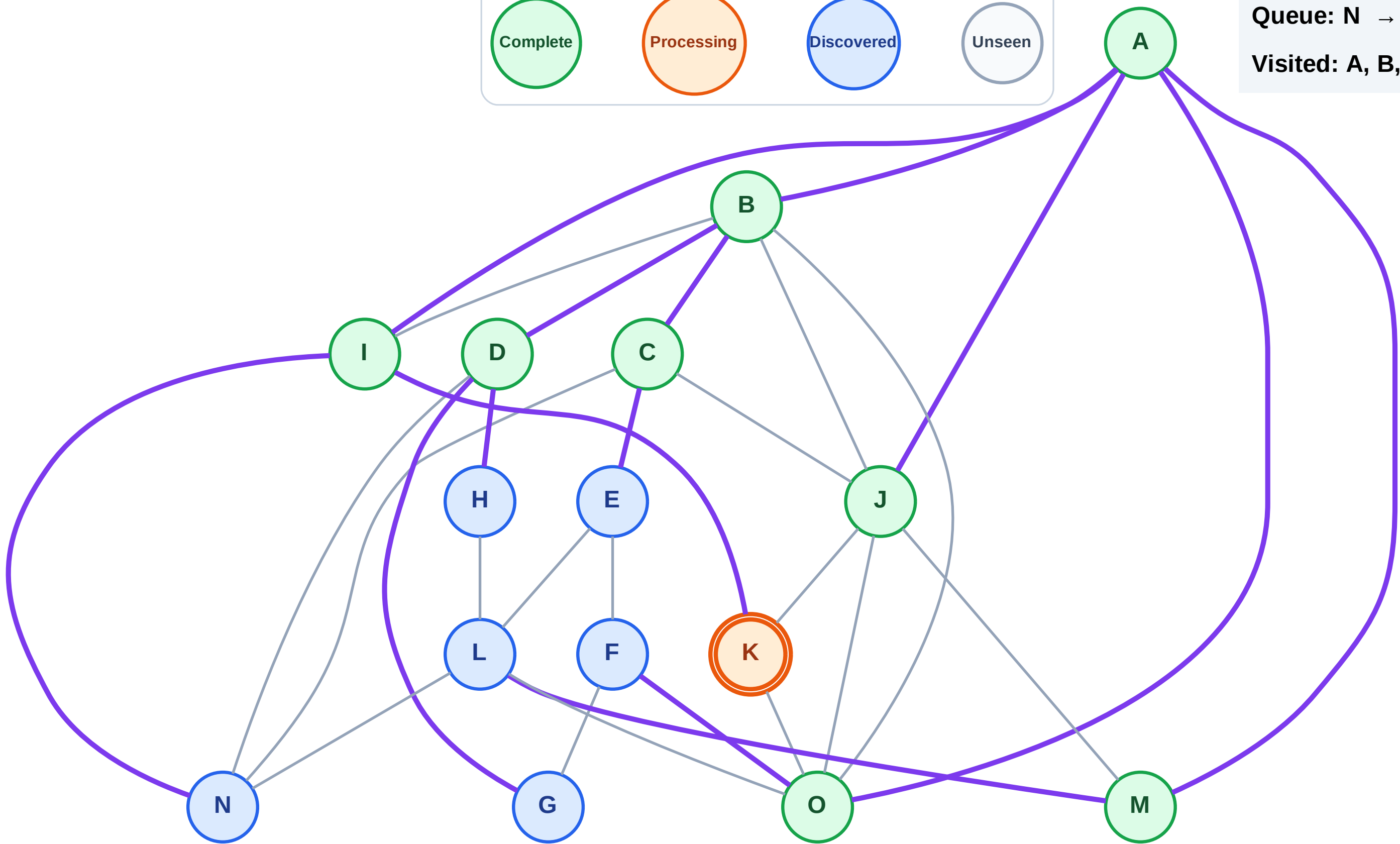
Step 18: Process node K

Legend



Queue: N → L → F → E → G → H

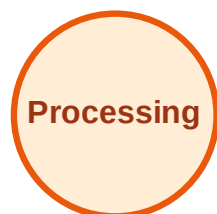
Visited: A, B, C, D, I, J, M, O



BFS Traversal

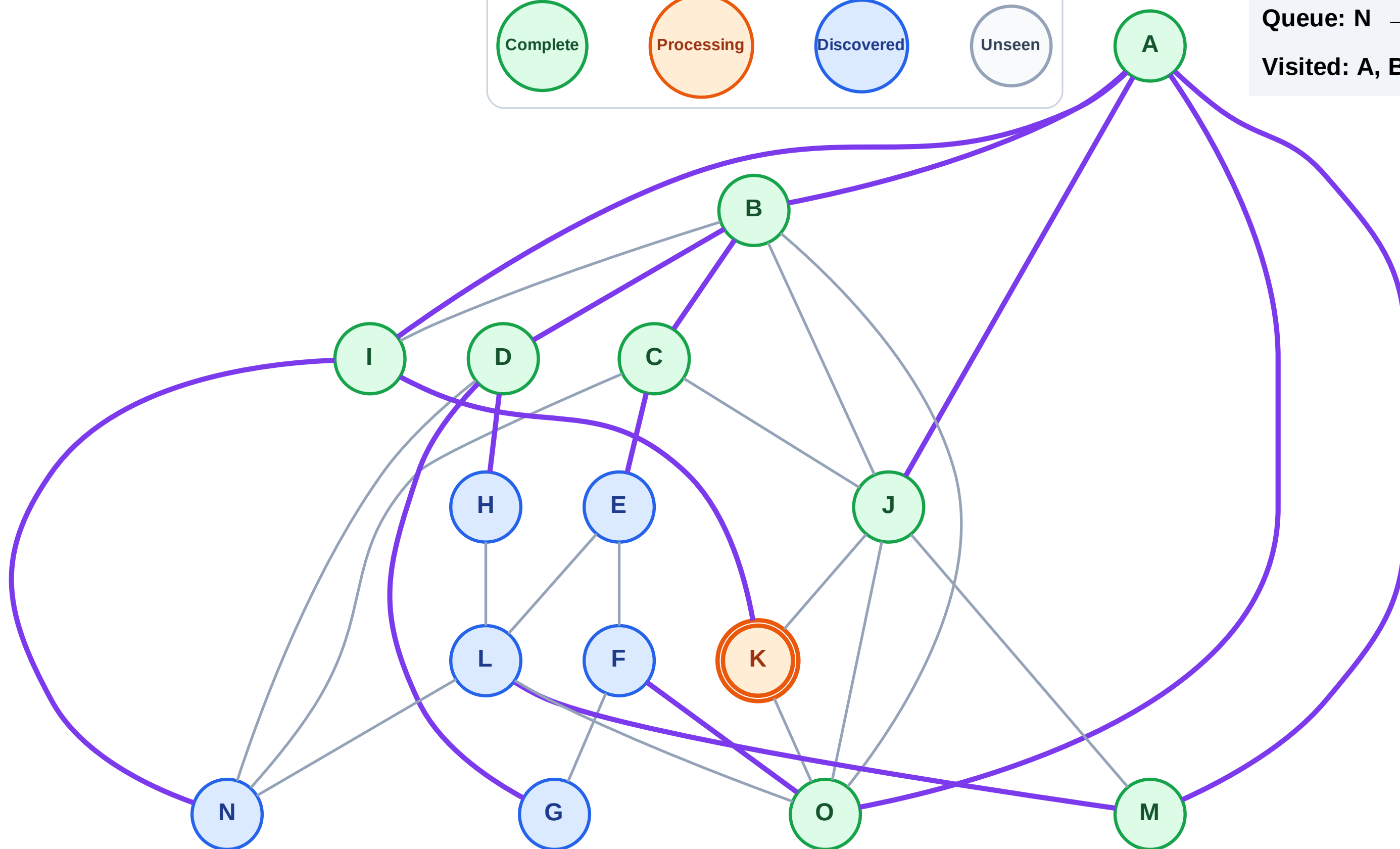
Step 19: TARGET NODE K REACHED! Path (2 hops): A → I → K

Legend



Queue: N → L → F → E → G → H

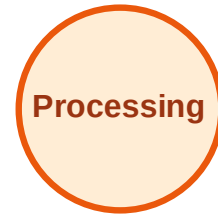
Visited: A, B, C, D, I, J, K, M, O



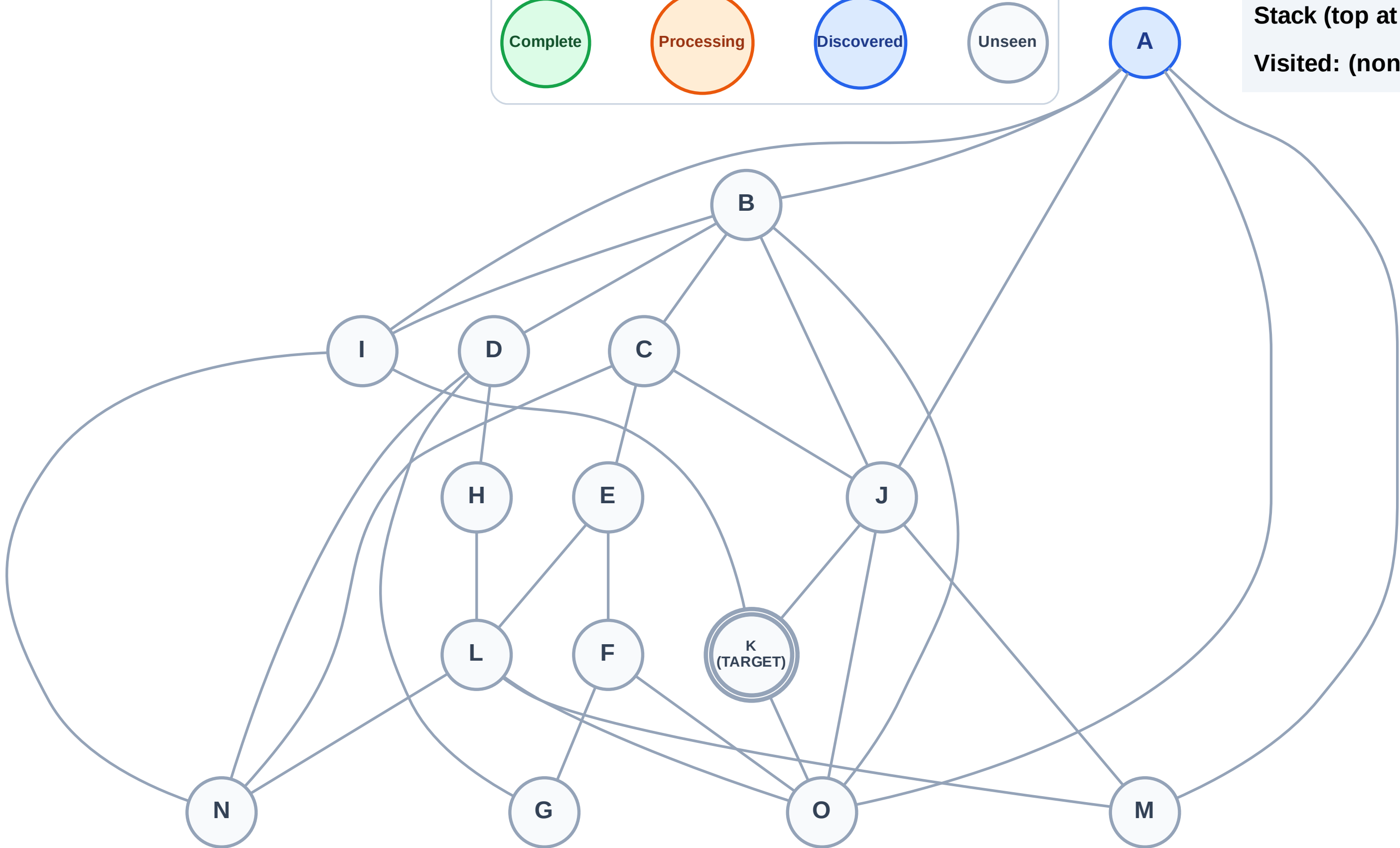
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

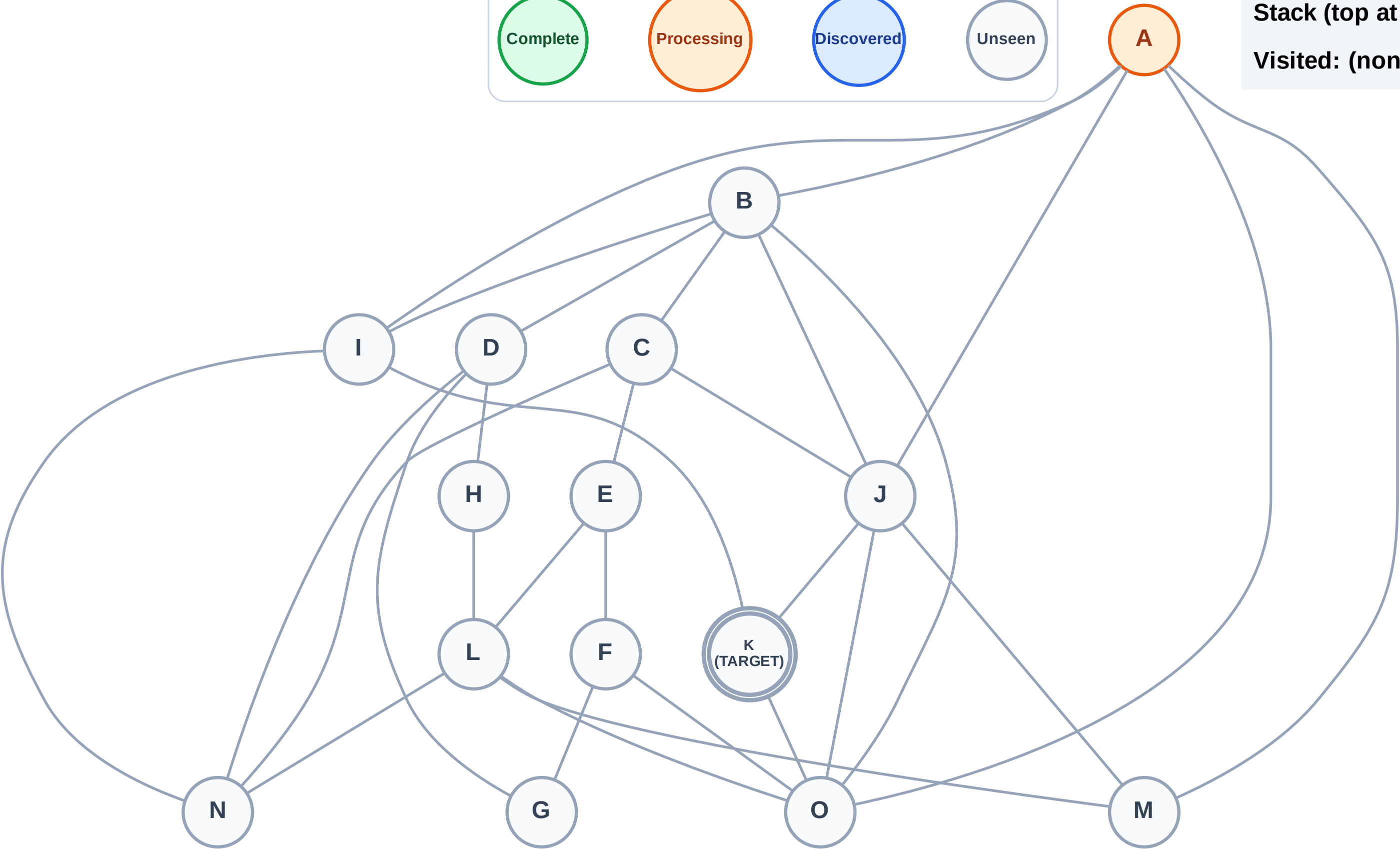
Complete

Processing

Discovered

Unseen

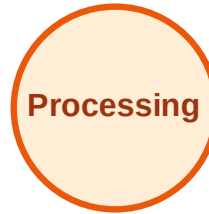
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

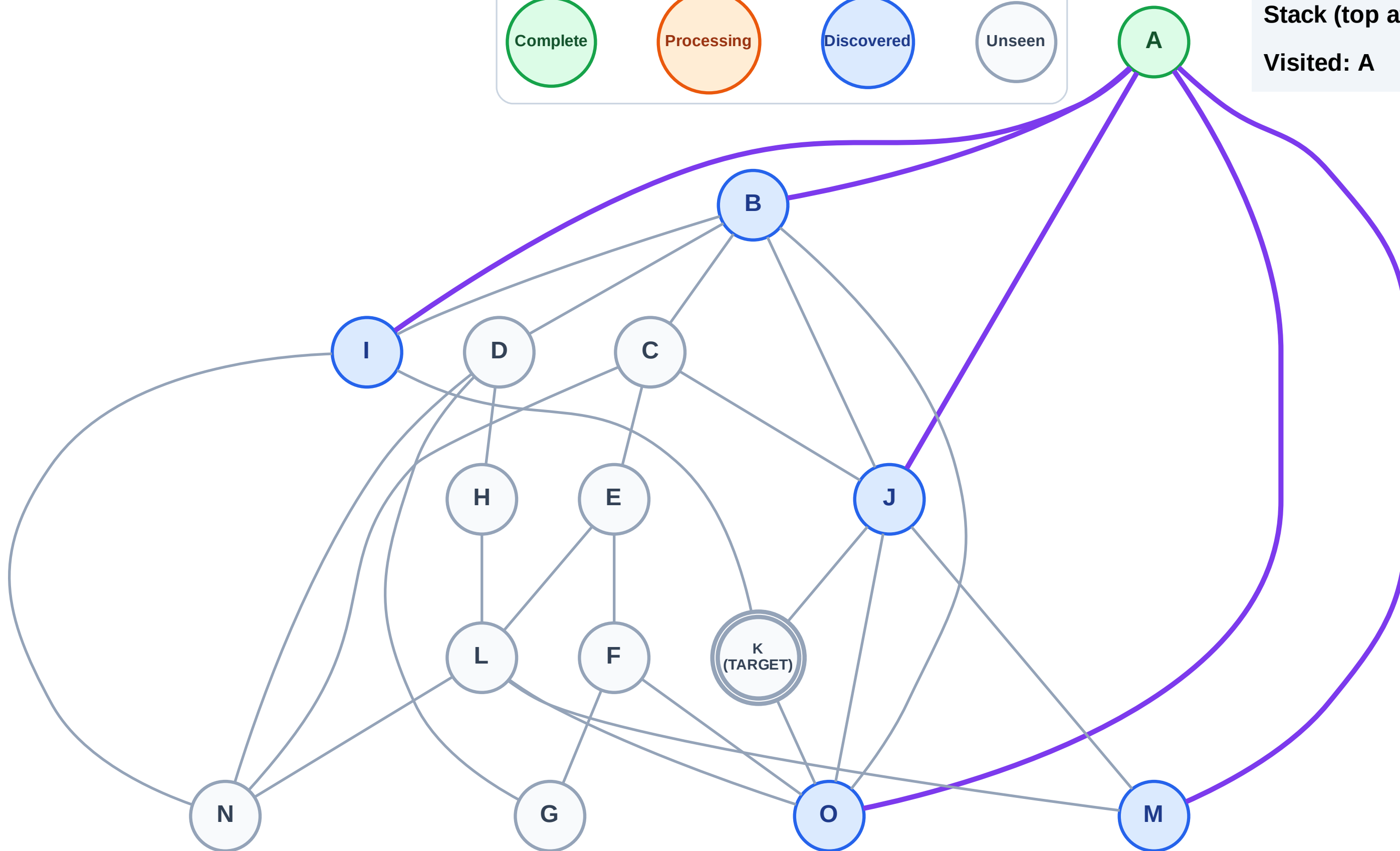
Step 3: Discover O, M, J, I, B from A

Legend



Stack (top at right): O | M | J | I | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

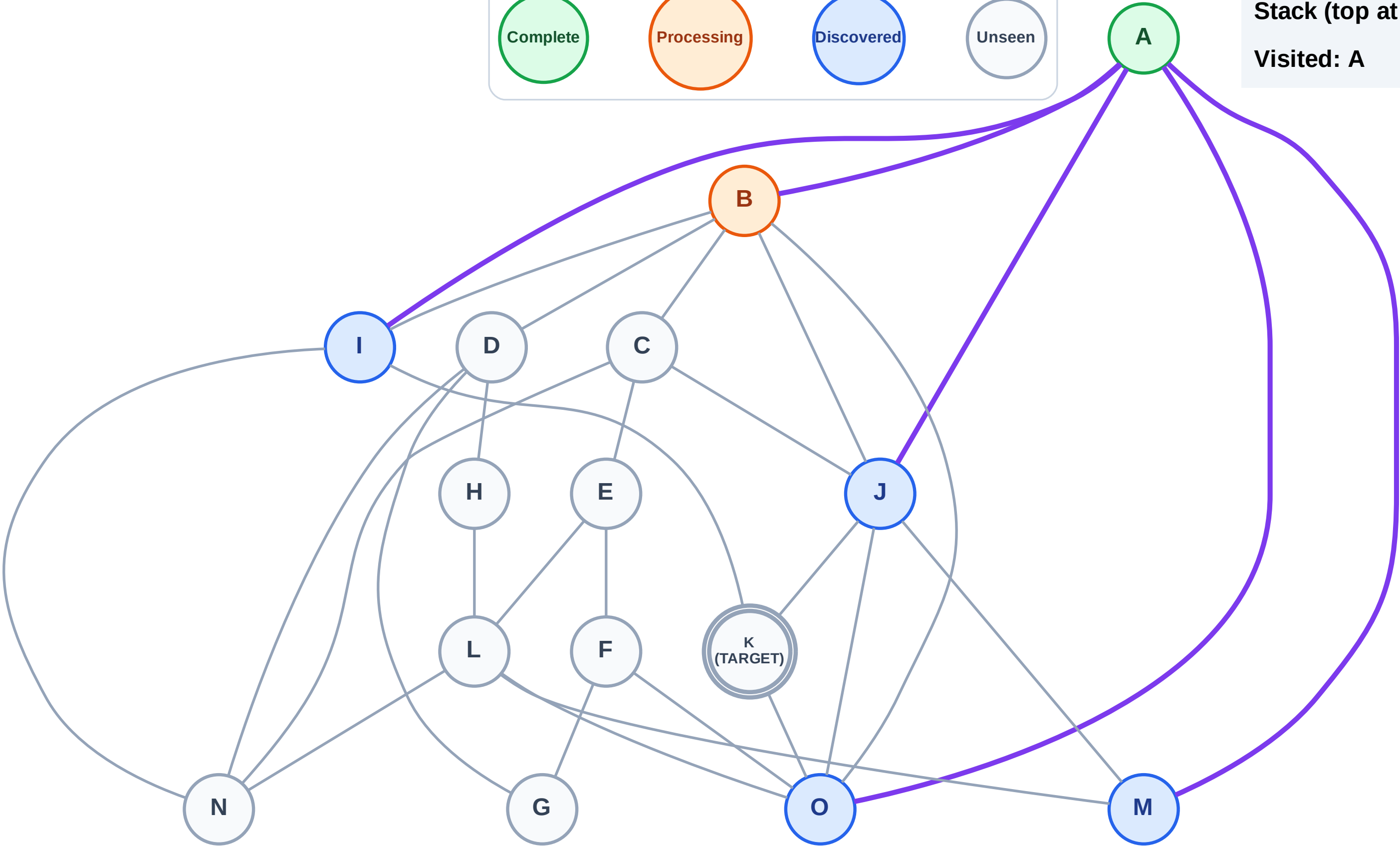
Complete

Processing

Discovered

Unseen

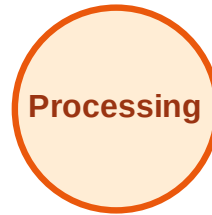
Stack (top at right): O | M | J | I
Visited: A



DFS Traversal

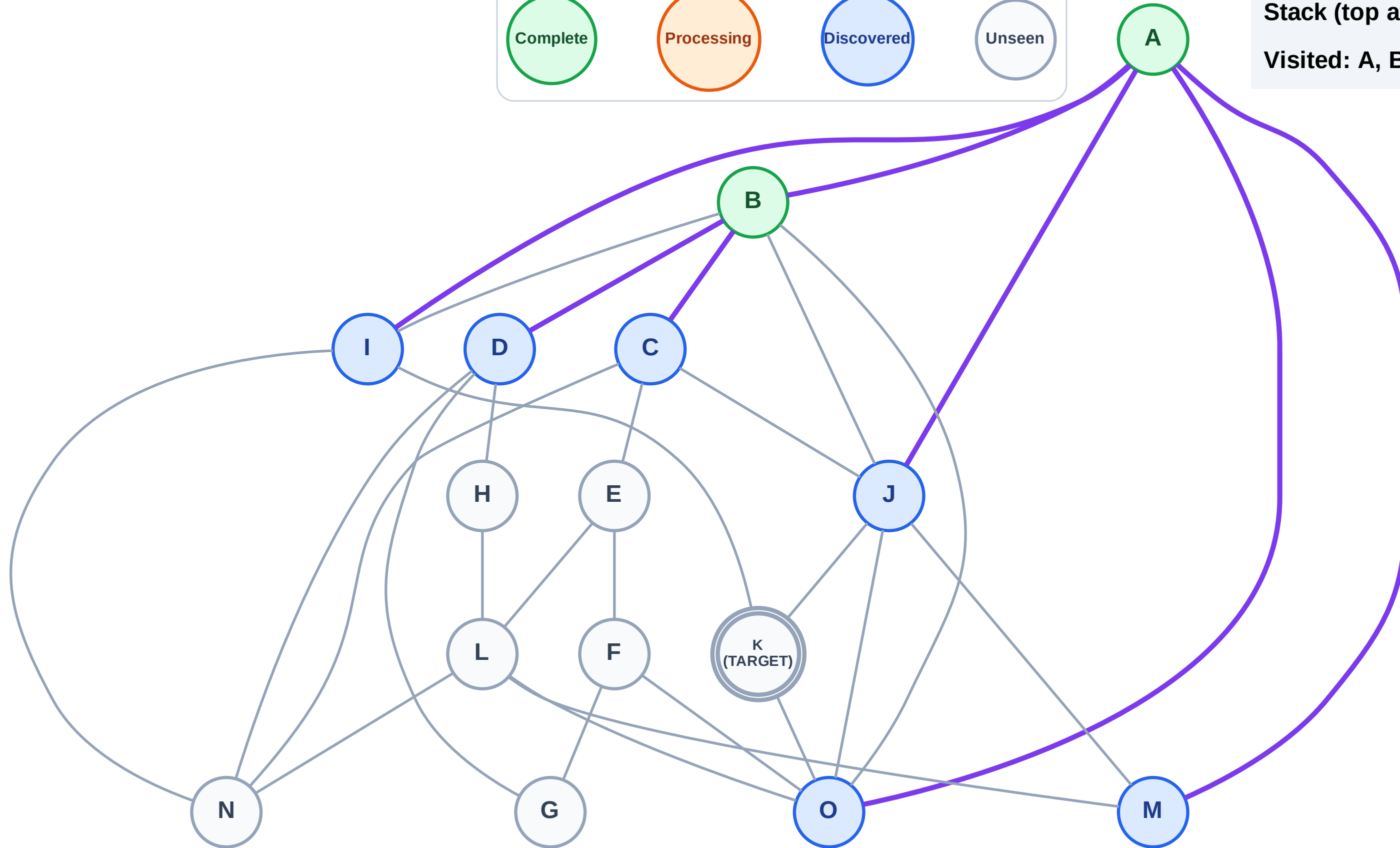
Step 5: Discover D, C from B

Legend



Stack (top at right): O | M | J | I | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

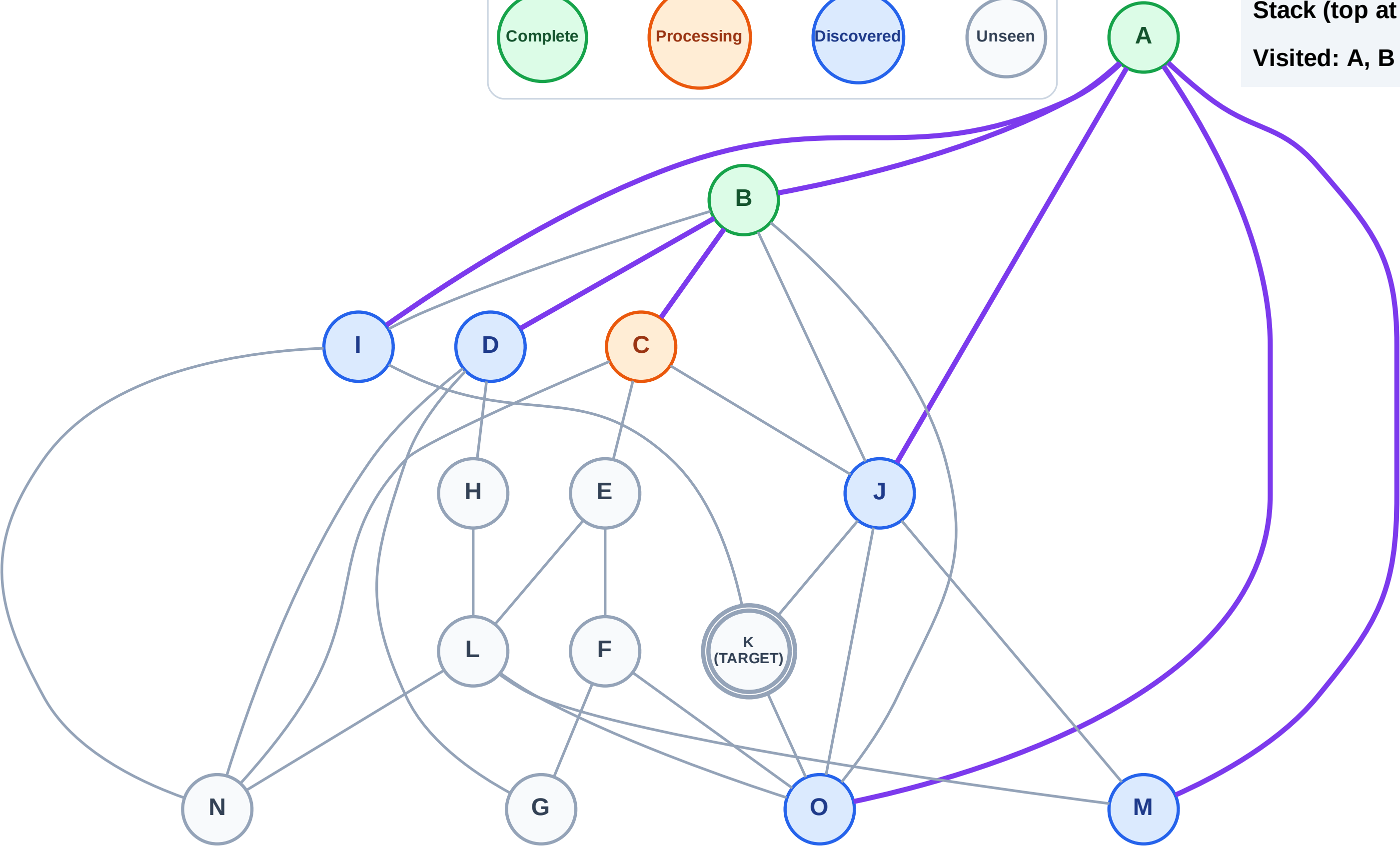
Processing

Discovered

Unseen

Stack (top at right): O | M | J | I | D

Visited: A, B



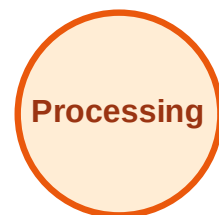
DFS Traversal

Step 7: Discover N, E from C

Legend



Complete



Processing



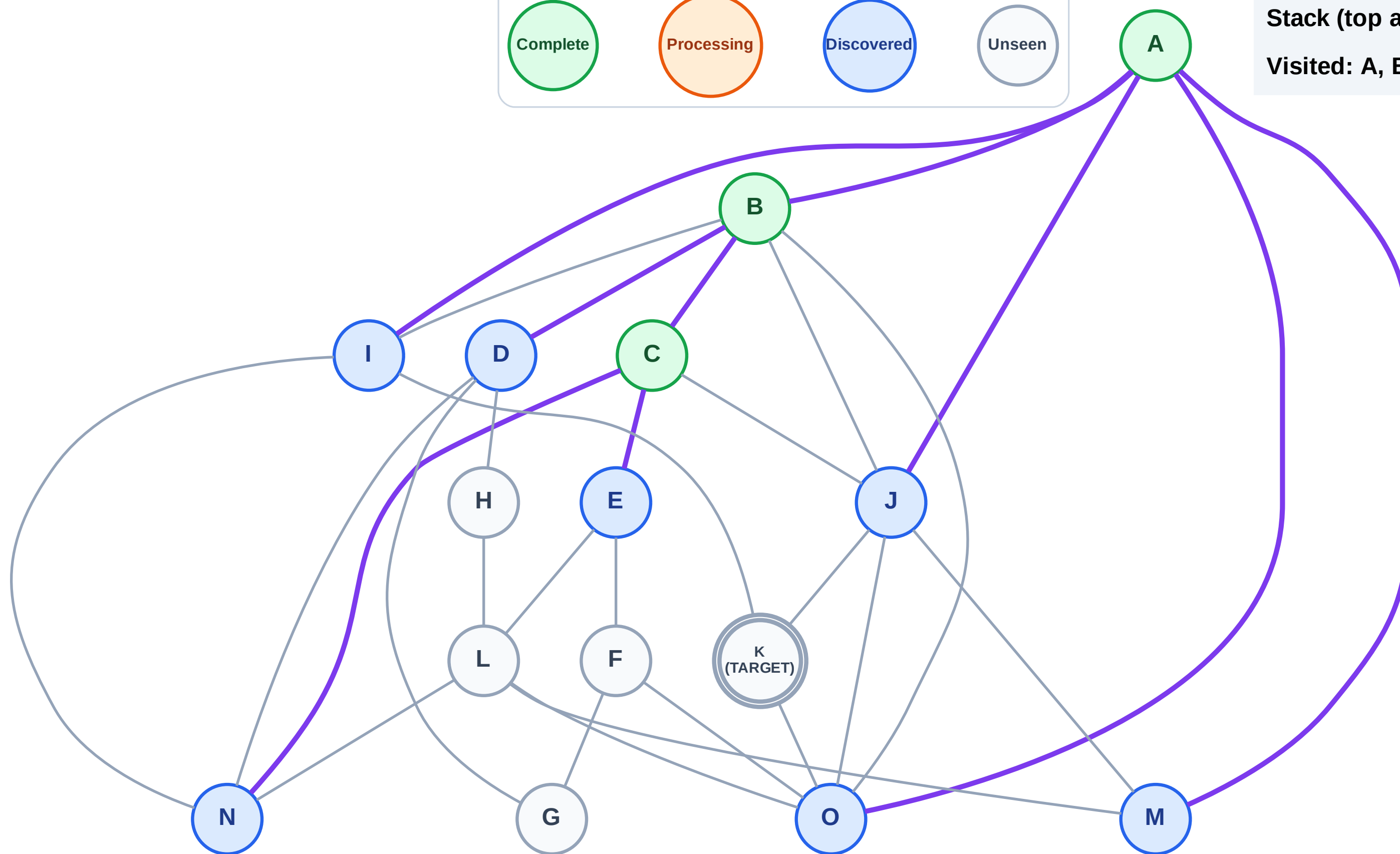
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | E

Visited: A, B, C



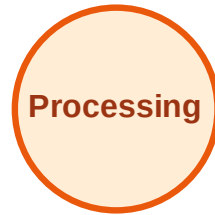
DFS Traversal

Step 8: Process node E

Legend



Complete



Processing



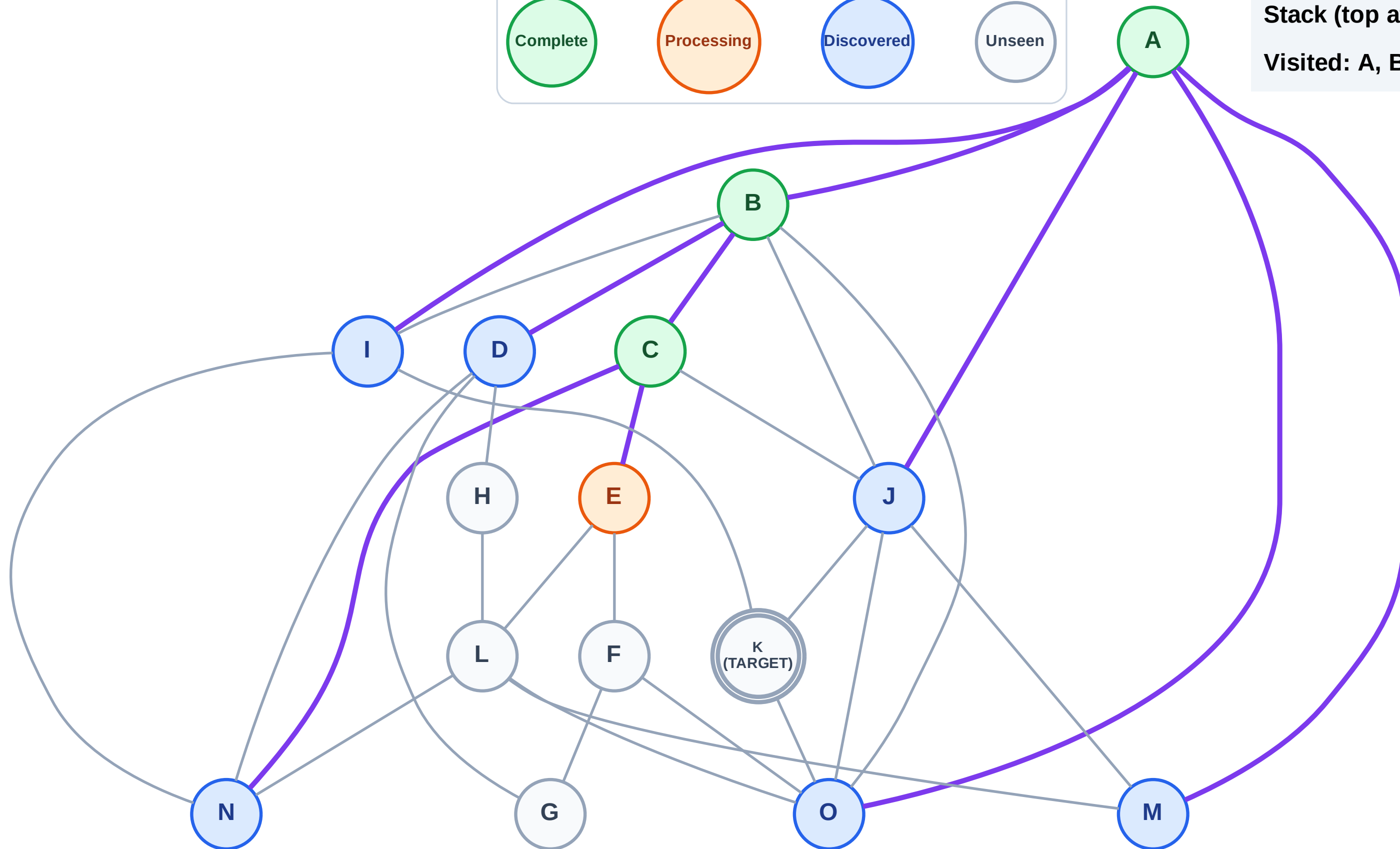
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N

Visited: A, B, C



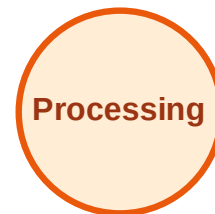
DFS Traversal

Step 9: Discover L, F from E

Legend



Complete



Processing



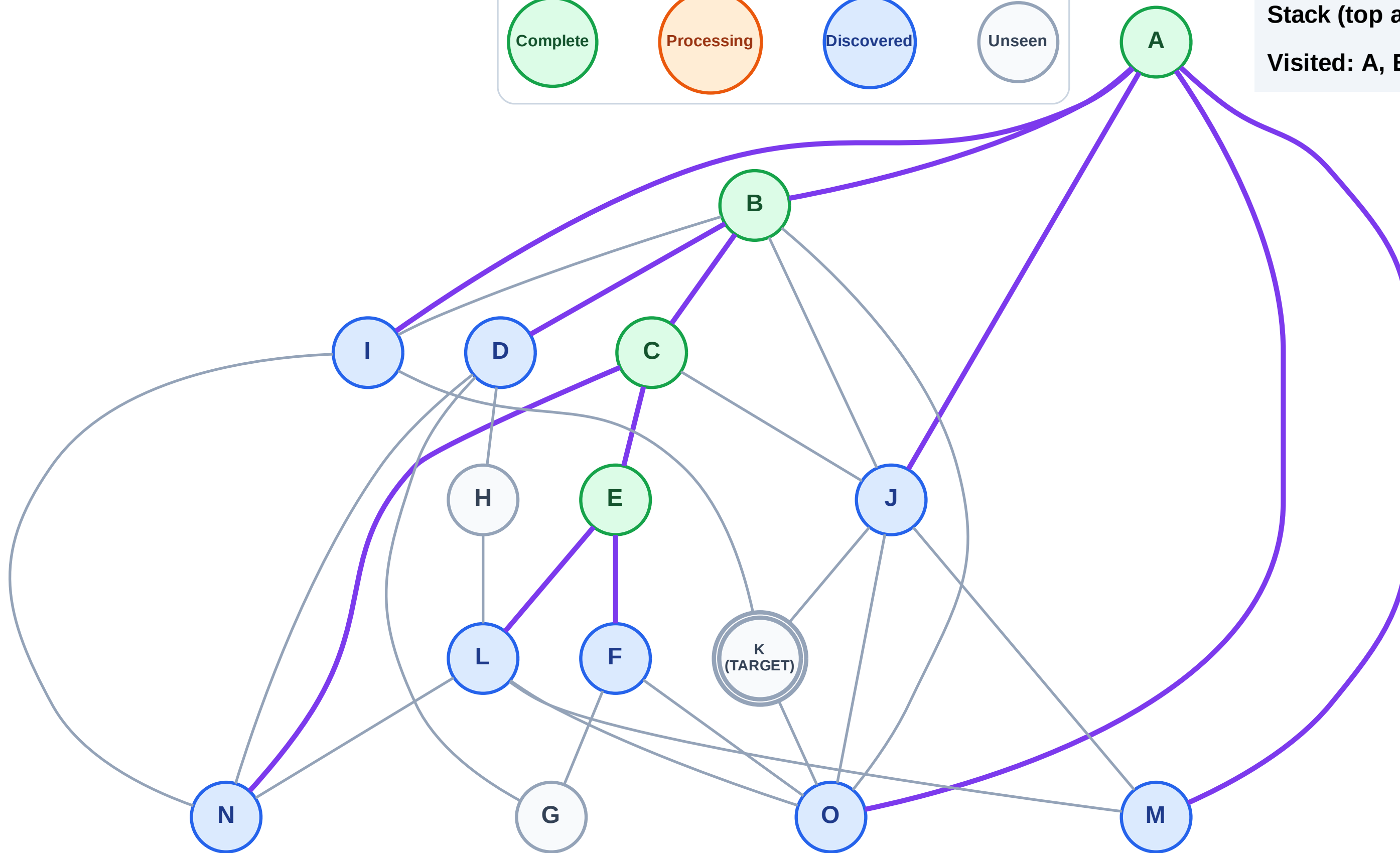
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L | F

Visited: A, B, C, E



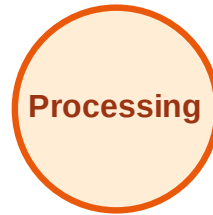
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



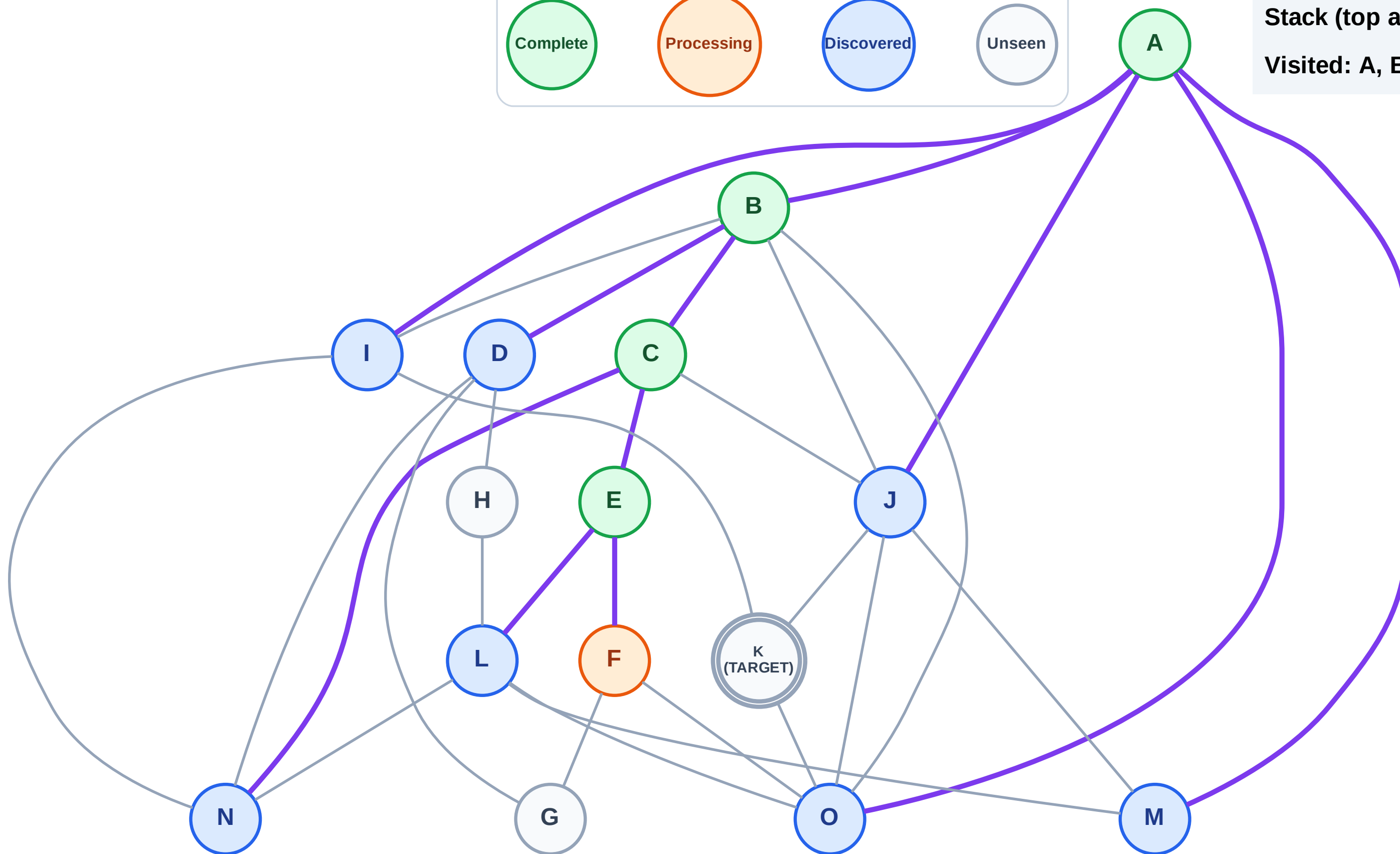
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L

Visited: A, B, C, E



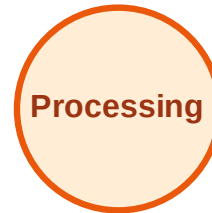
DFS Traversal

Step 11: Discover G from F

Legend



Complete



Processing



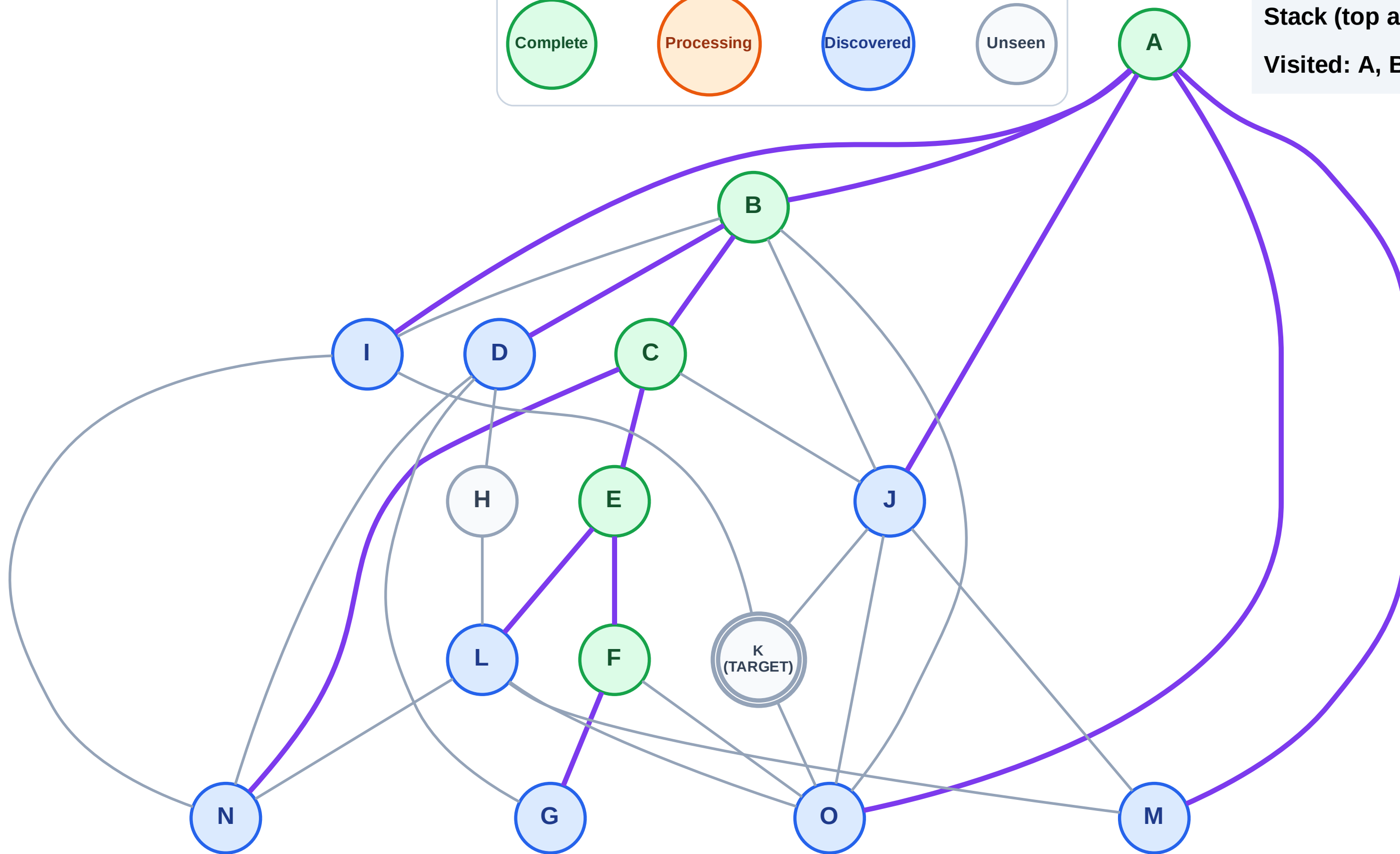
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L | G

Visited: A, B, C, E, F



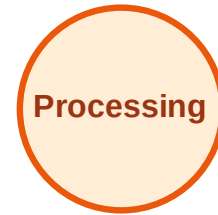
DFS Traversal

Step 12: Process node G

Legend



Complete



Processing



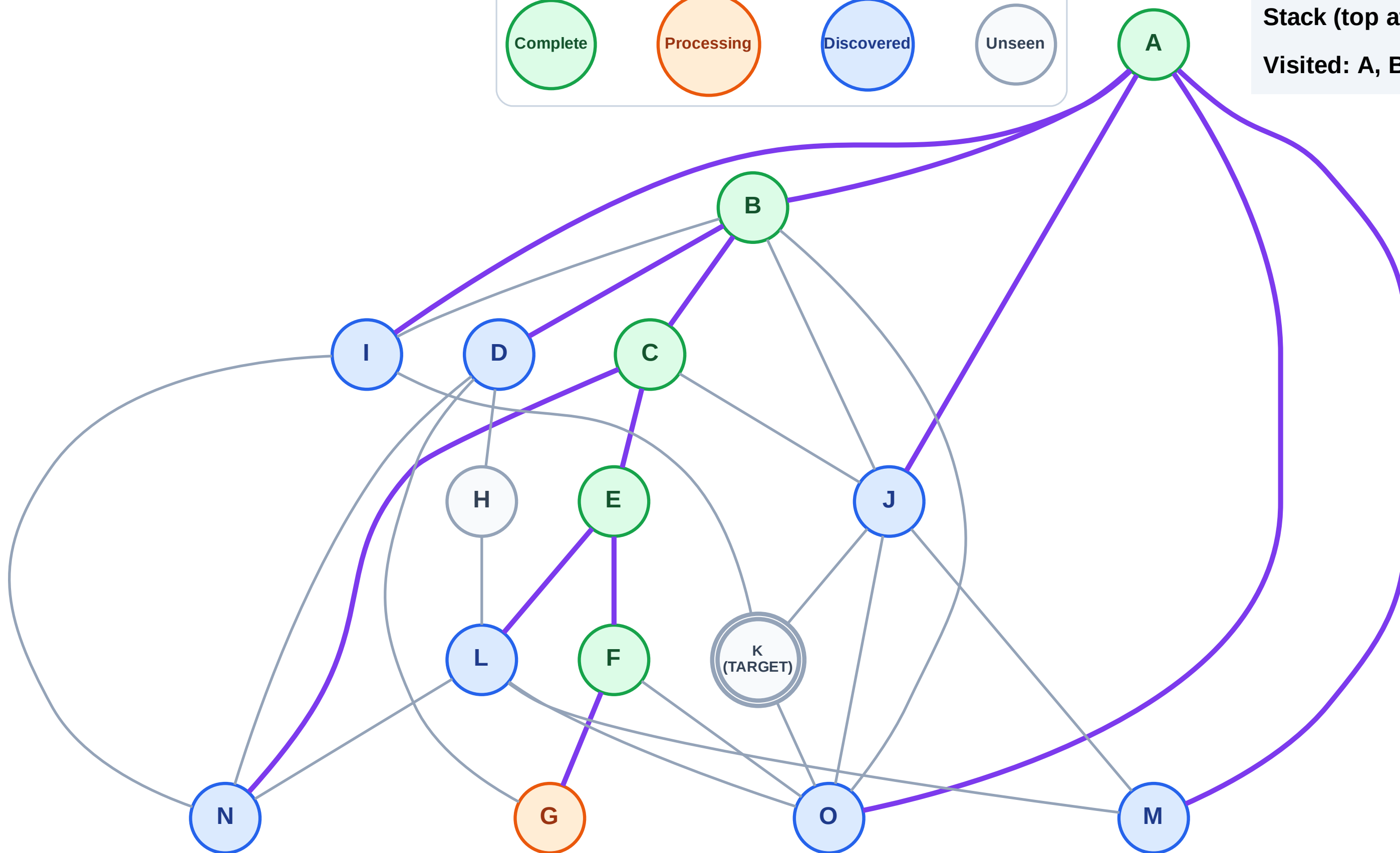
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L

Visited: A, B, C, E, F



Step 13: No new neighbors from G

Step 13: No new neighbors from G

Legend



Complete Processing Discovered Unseen

Complete

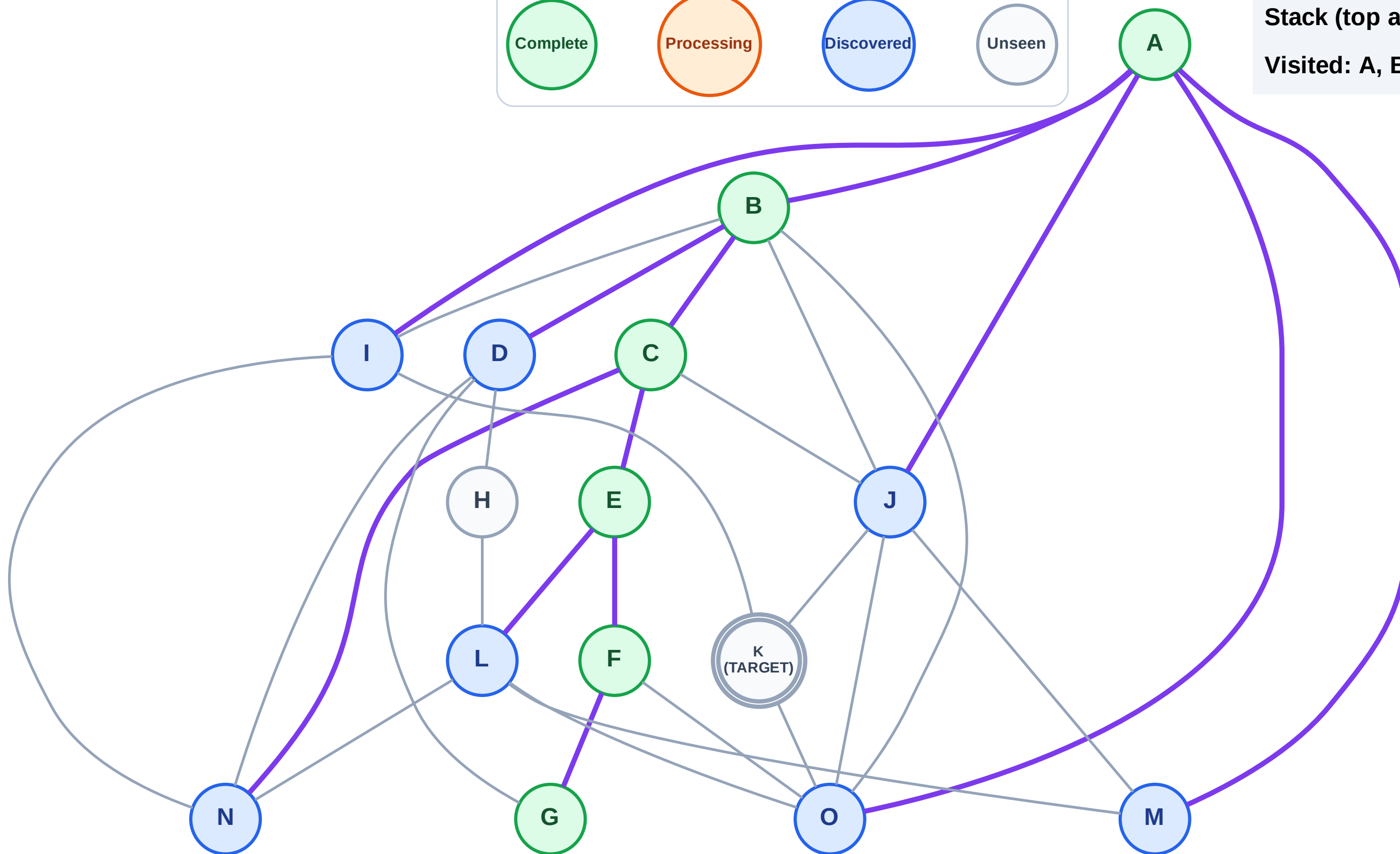
Processing

Discovered

Unseen

Stack (top at right): O | M | J | I | D | N | L

Visited: A, B, C, E, F, G



DFS Traversal

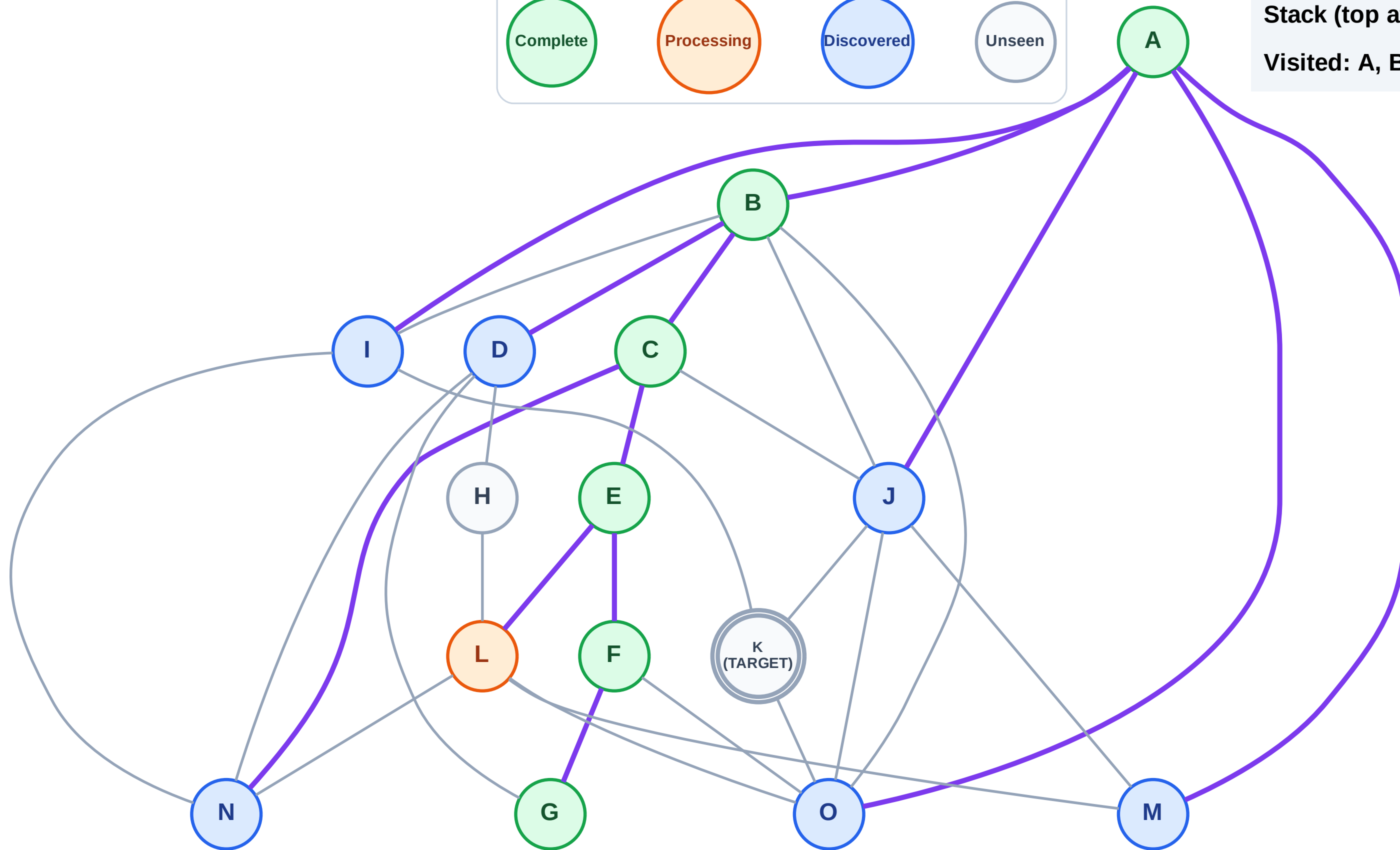
Step 14: Process node L

Legend



Stack (top at right): O | M | J | I | D | N

Visited: A, B, C, E, F, G



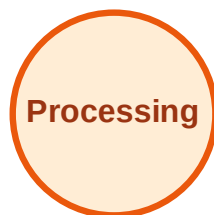
DFS Traversal

Step 15: Discover H from L

Legend



Complete



Processing



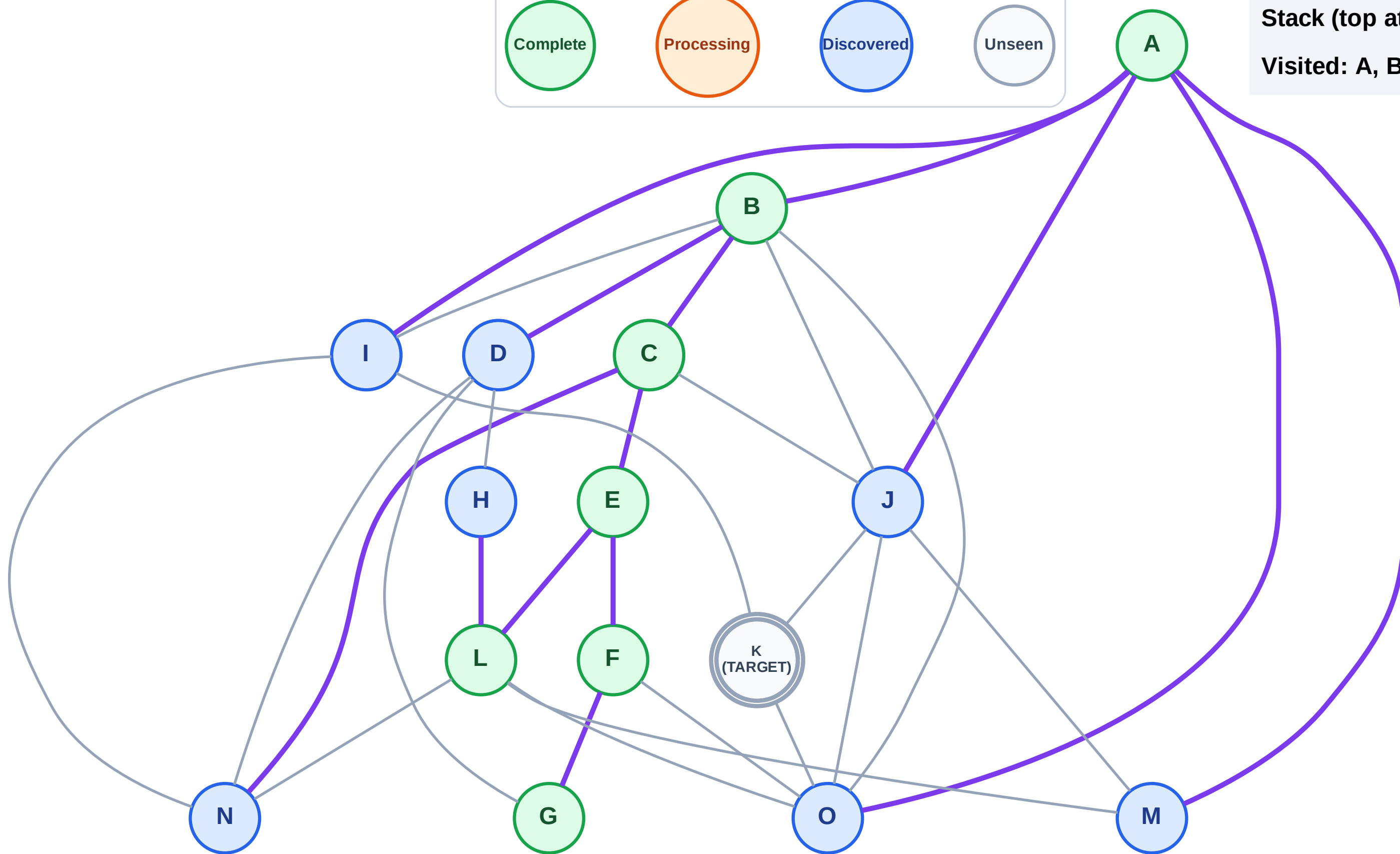
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | H

Visited: A, B, C, E, F, G, L



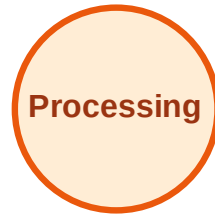
DFS Traversal

Step 16: Process node H

Legend



Complete



Processing



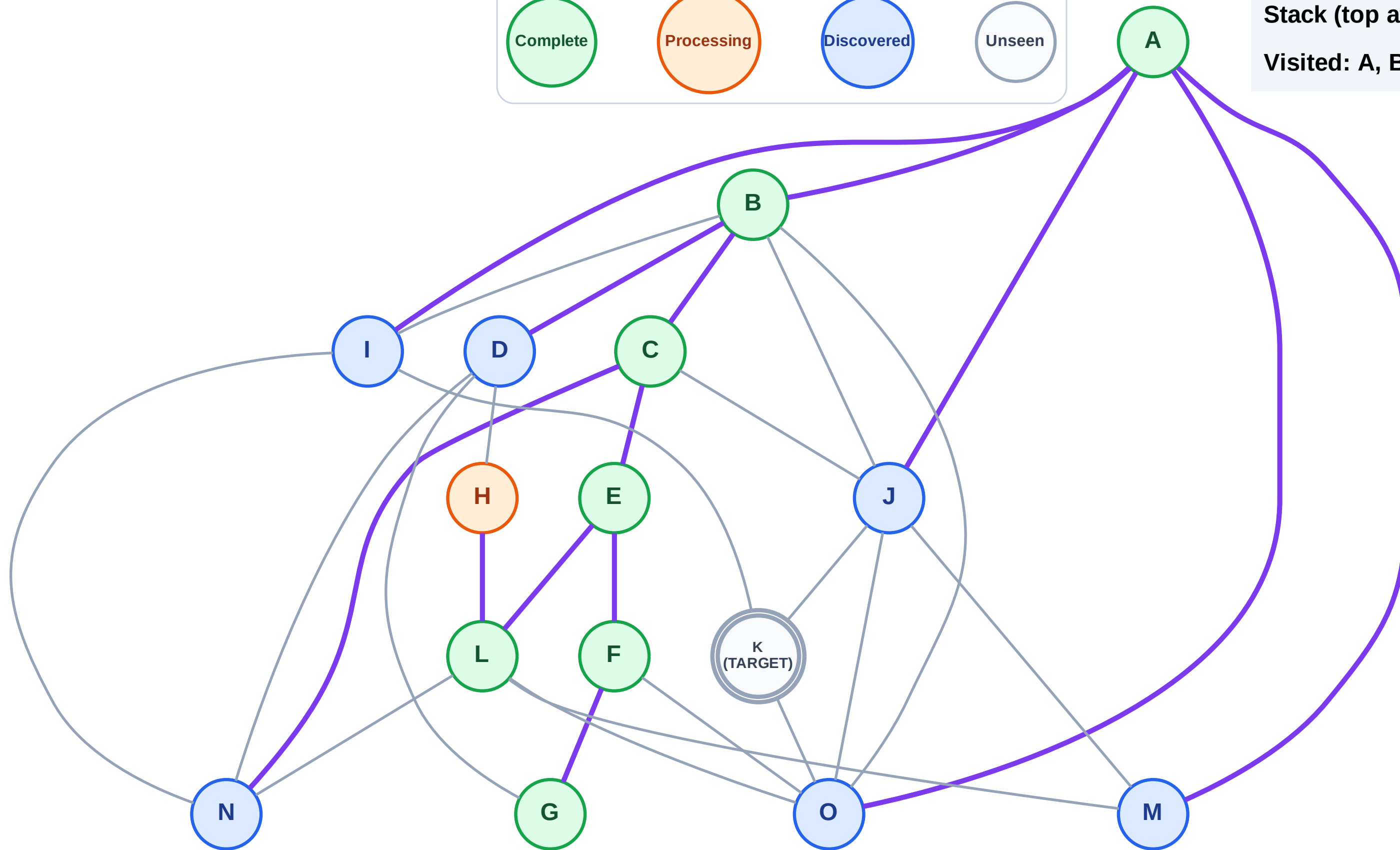
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N

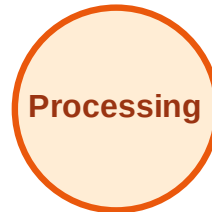
Visited: A, B, C, E, F, G, L



DFS Traversal

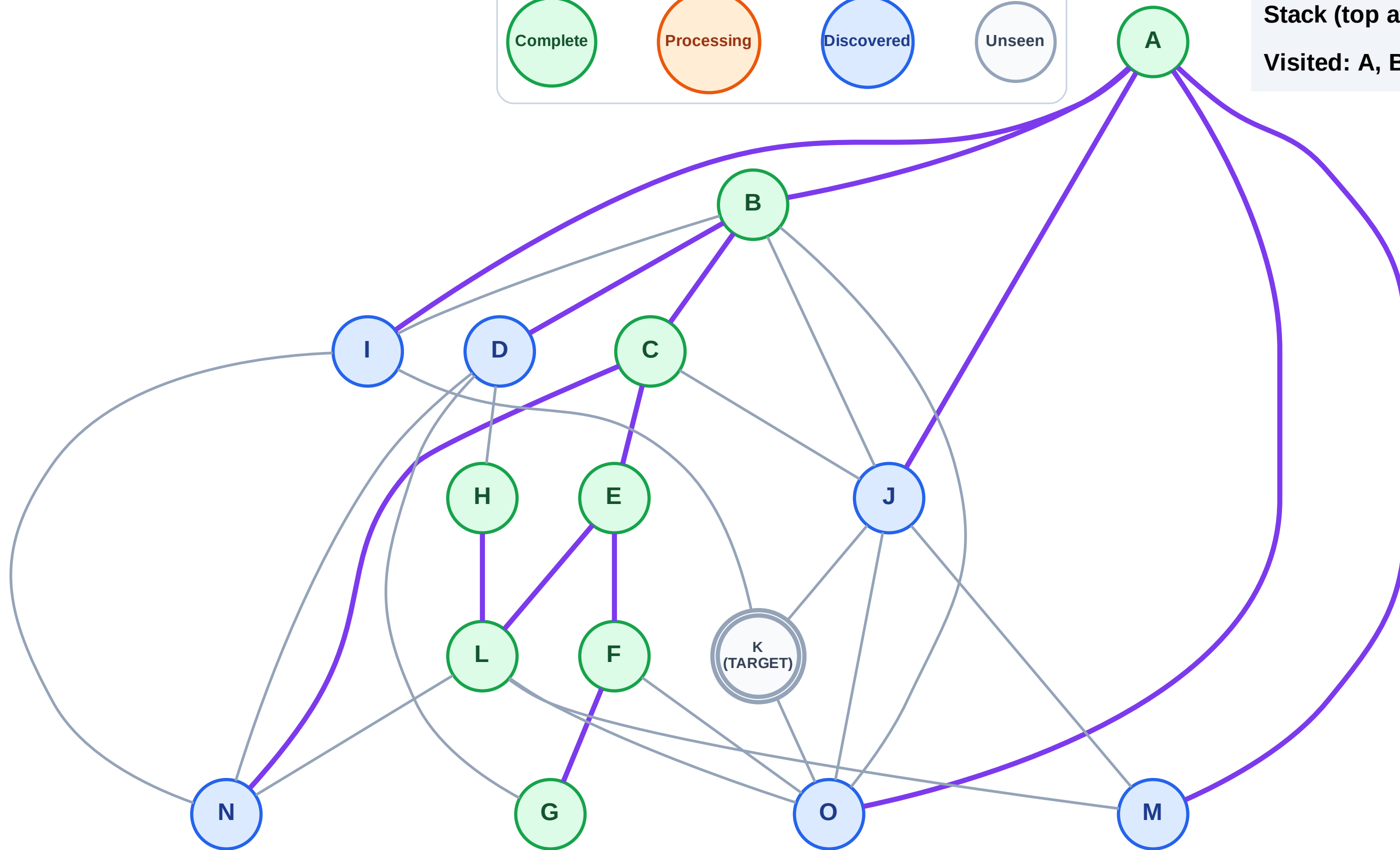
Step 17: No new neighbors from H

Legend



Stack (top at right): O | M | J | I | D | N

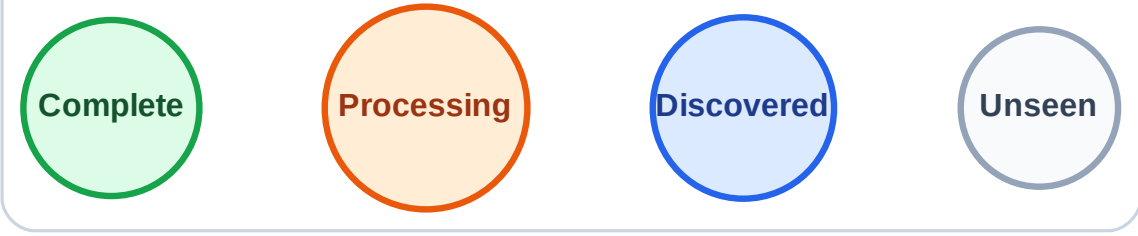
Visited: A, B, C, E, F, G, H, L



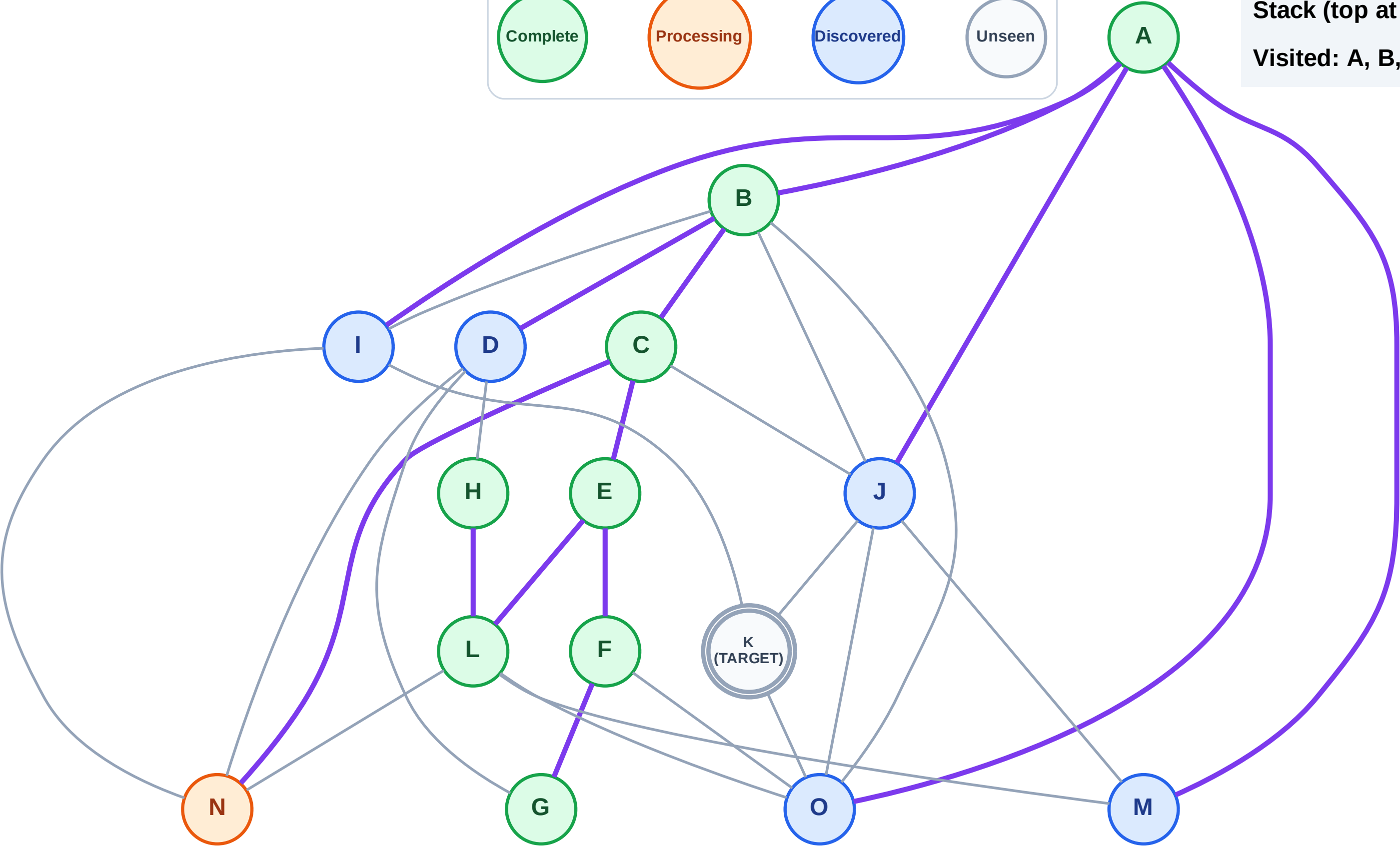
DFS Traversal

Step 18: Process node N

Legend



Stack (top at right): O | M | J | I | D
Visited: A, B, C, E, F, G, H, L



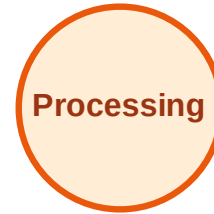
DFS Traversal

Step 19: No new neighbors from N

Legend



Complete



Processing



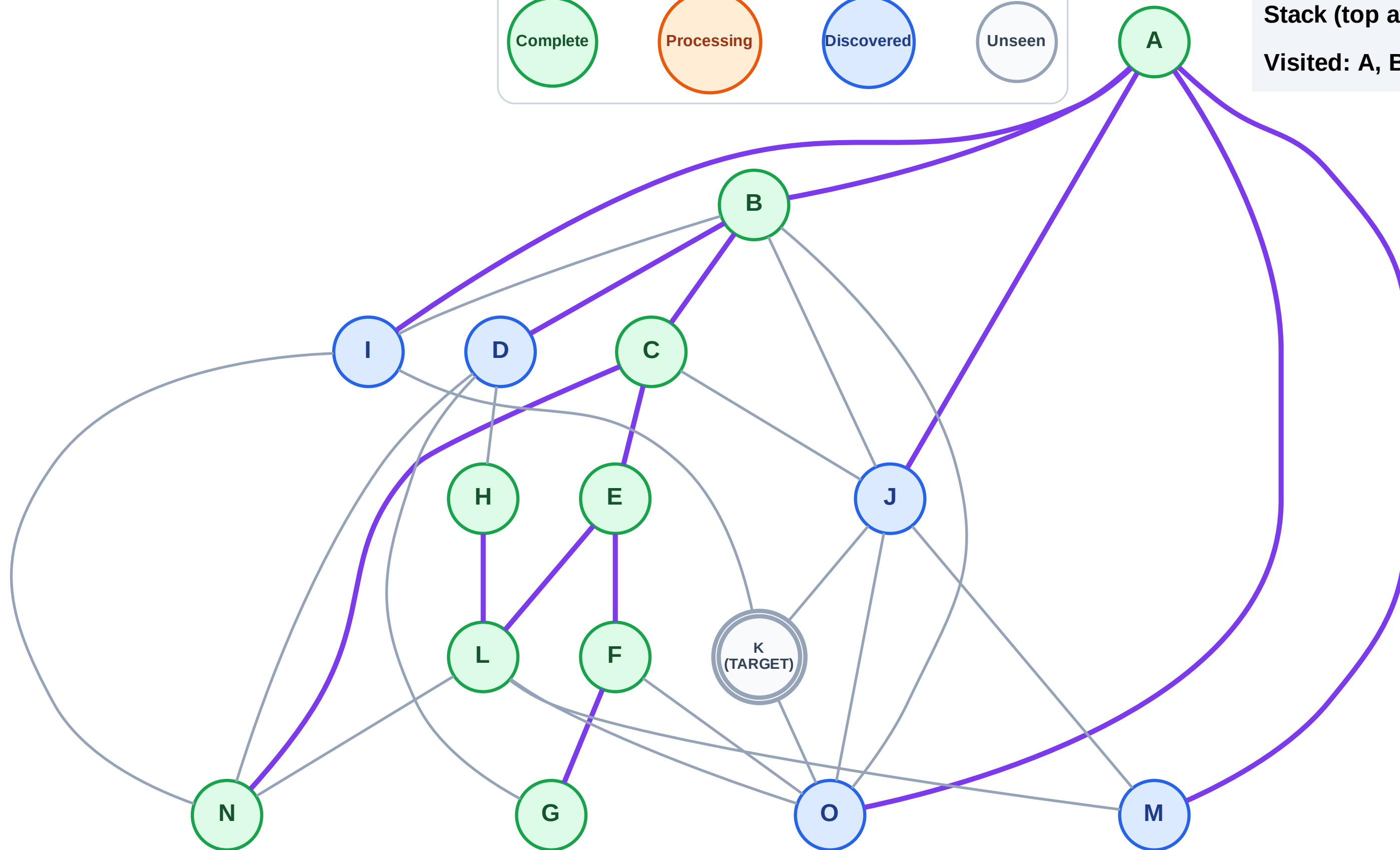
Discovered



Unseen

Stack (top at right): O | M | J | I | D

Visited: A, B, C, E, F, G, H, L, N



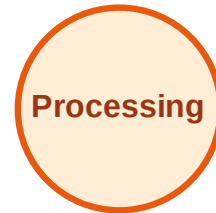
DFS Traversal

Step 20: Process node D

Legend



Complete



Processing



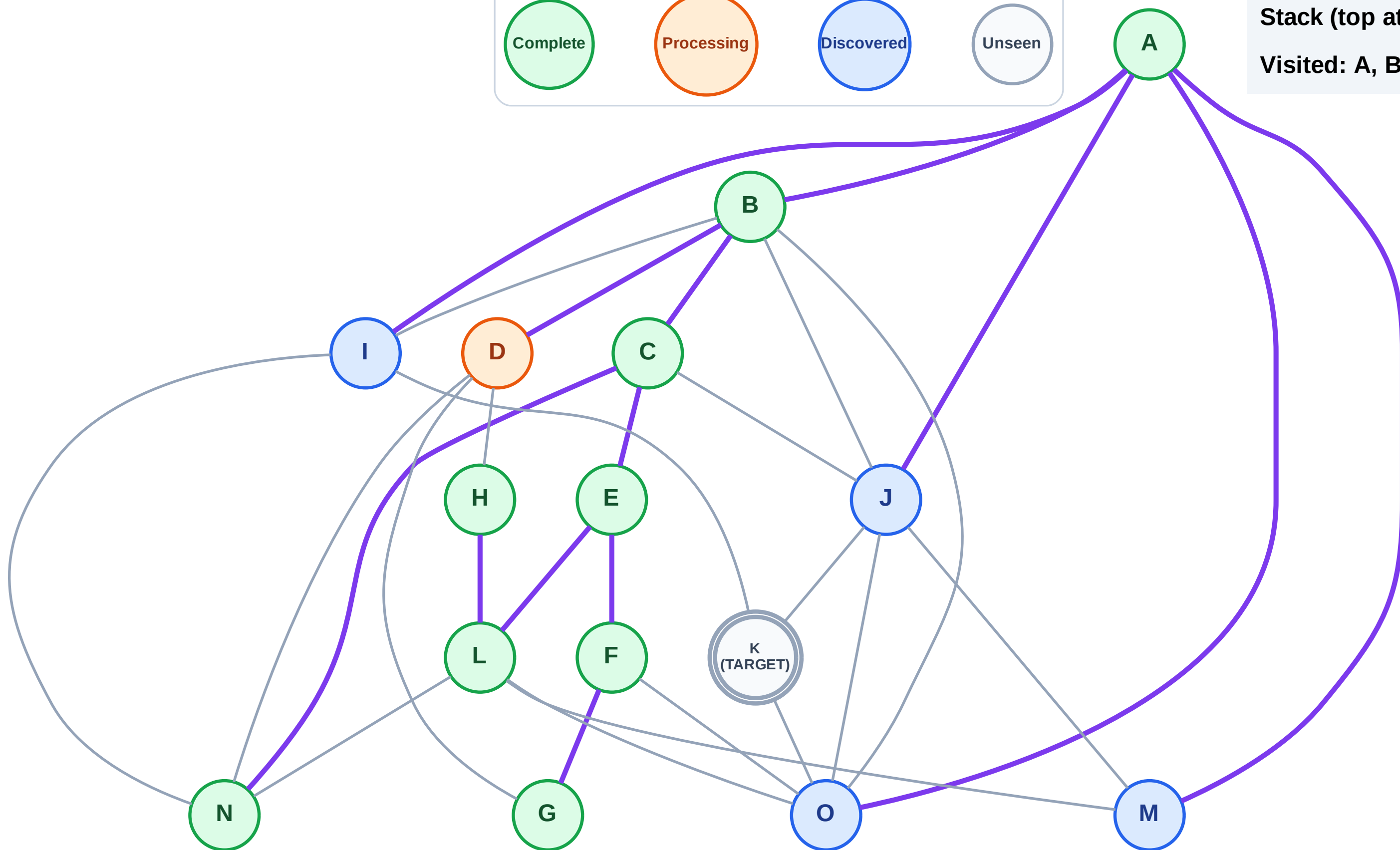
Discovered



Unseen

Stack (top at right): O | M | J | I

Visited: A, B, C, E, F, G, H, L, N



DFS Traversal

Step 21: No new neighbors from D

Legend

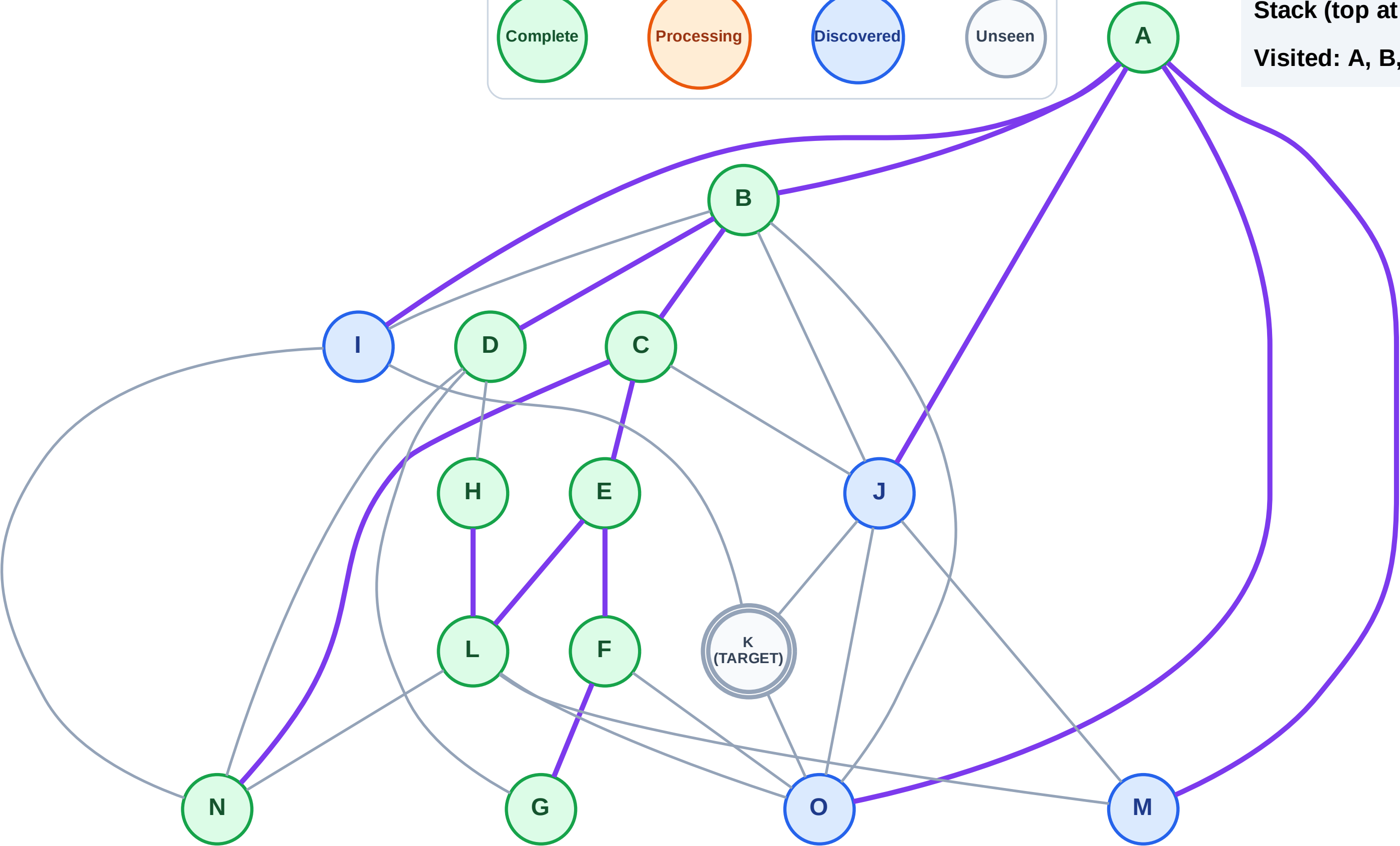
Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | J | I
Visited: A, B, C, D, E, F, G, H, L, N



DFS Traversal

Step 22: Process node I

Legend

Complete

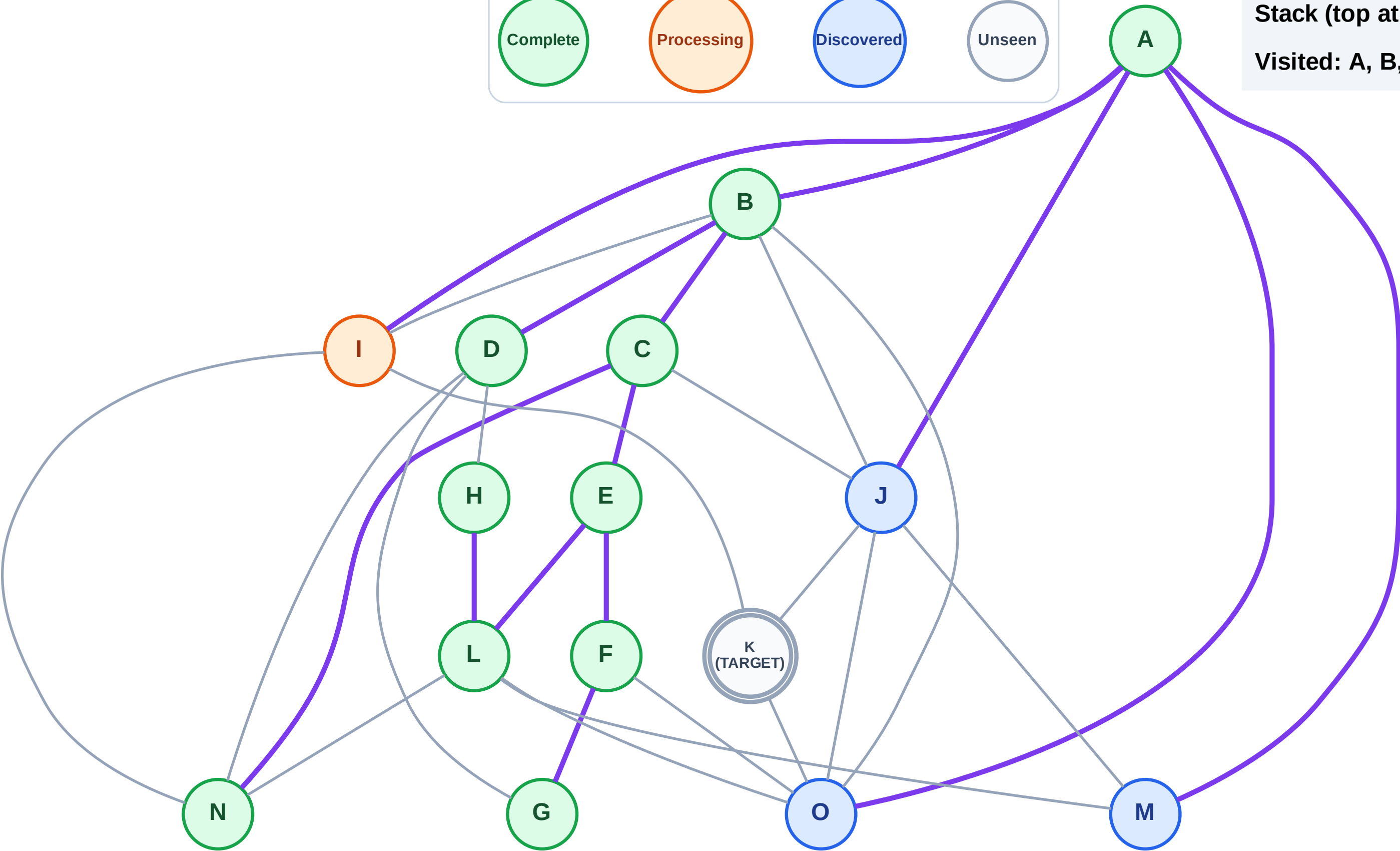
Processing

Discovered

Unseen

Stack (top at right): O | M | J

Visited: A, B, C, D, E, F, G, H, L, N



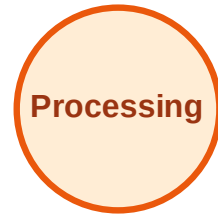
DFS Traversal

Step 23: Discover K from I

Legend



Complete



Processing



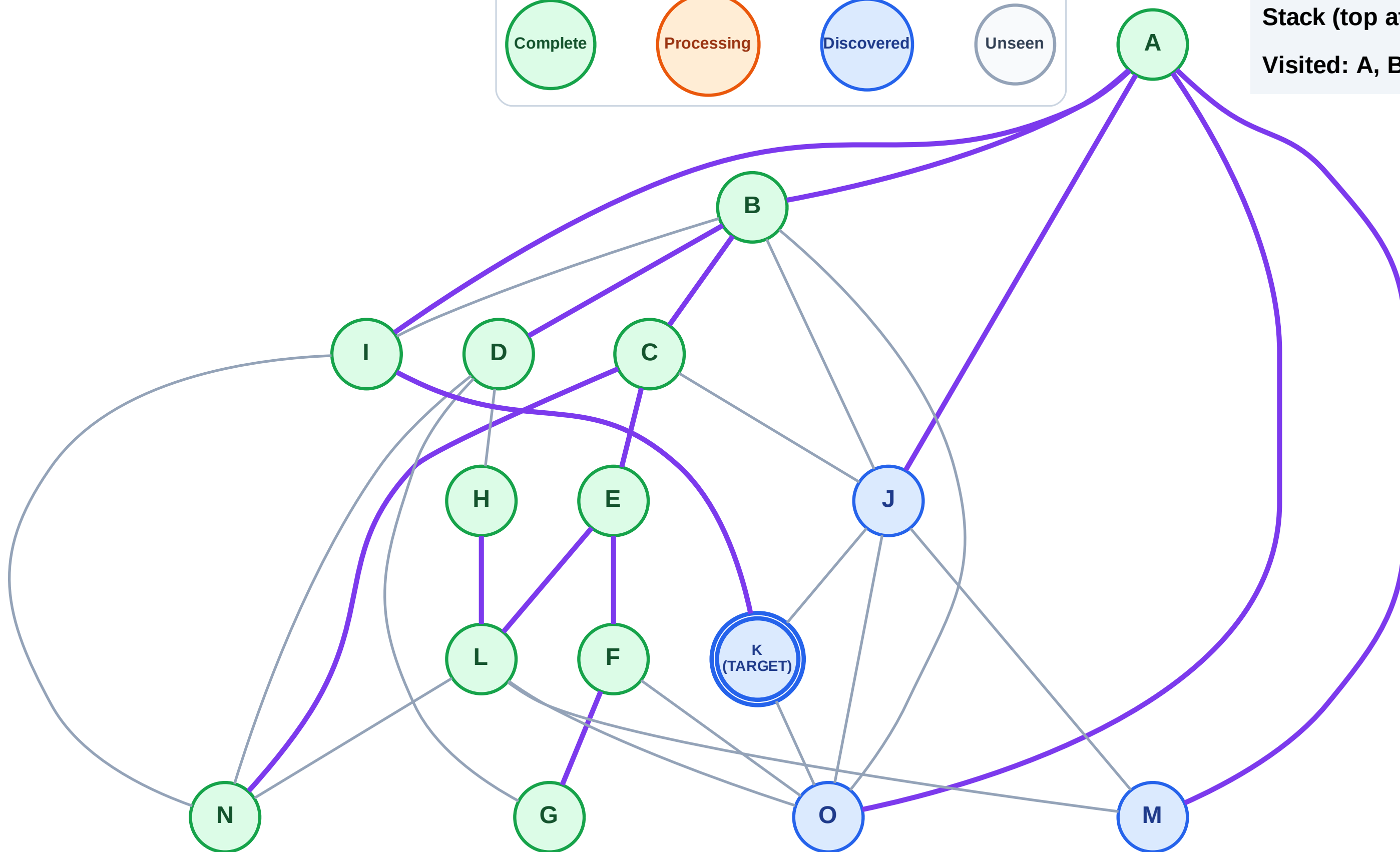
Discovered



Unseen

Stack (top at right): O | M | J | K

Visited: A, B, C, D, E, F, G, H, I, L, N



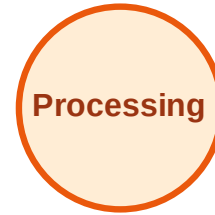
DFS Traversal

Step 24: Process node K

Legend



Complete



Processing



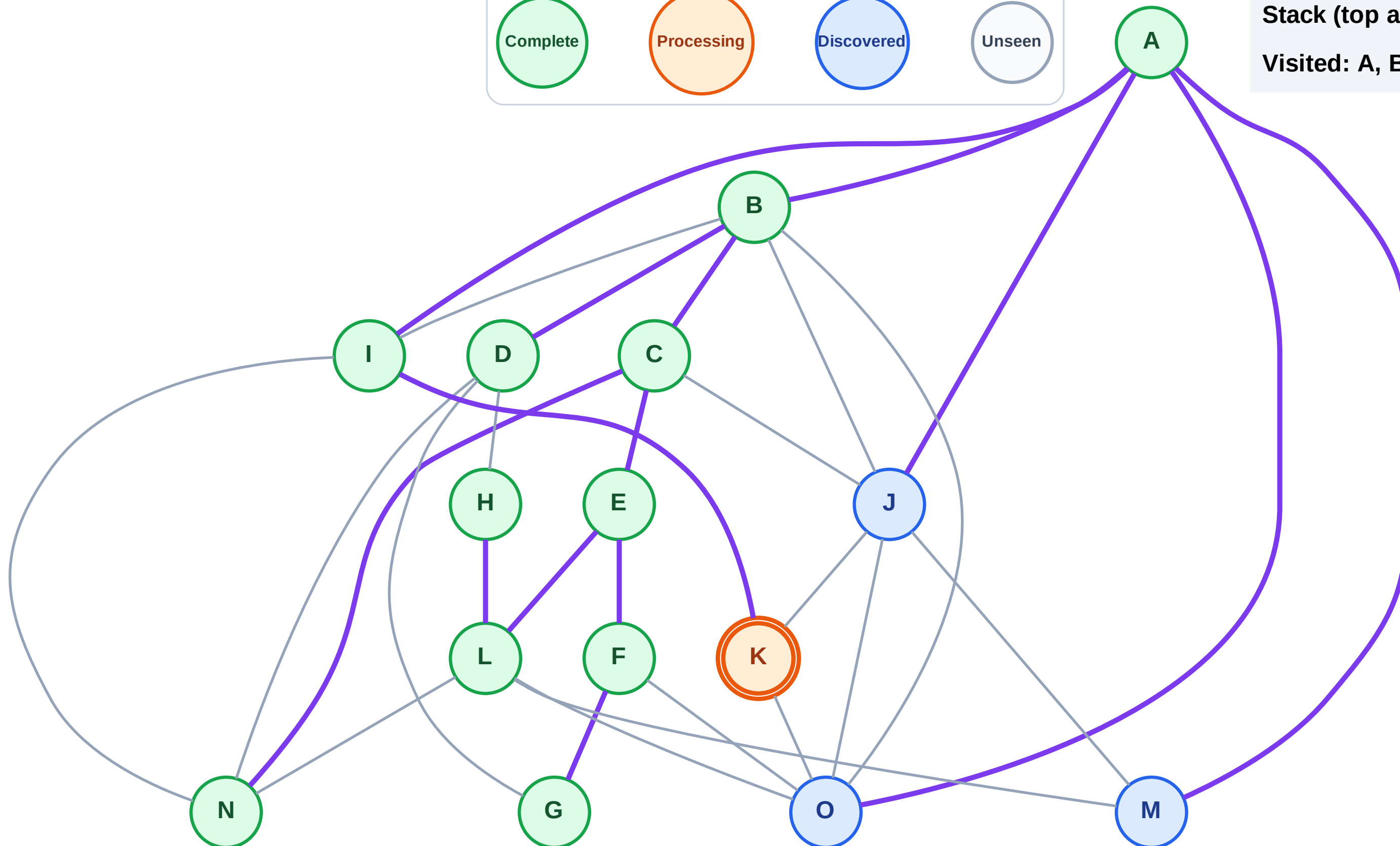
Discovered



Unseen

Stack (top at right): O | M | J

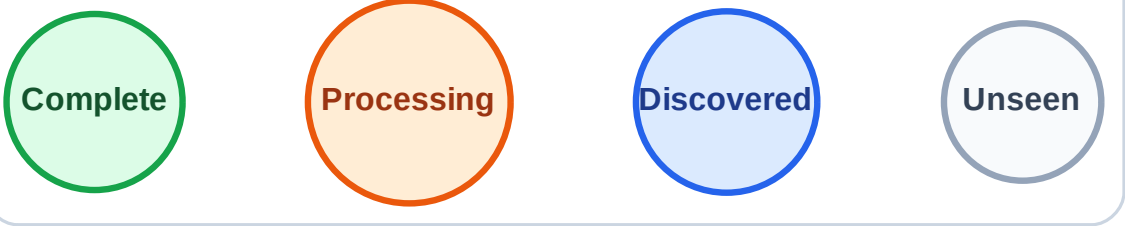
Visited: A, B, C, D, E, F, G, H, I, L, N



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → I → K

Legend



Stack (top at right): O | M | J

Visited: A, B, C, D, E, F, G, H, I, K, L, N

